This demo contains:

- **Data Exploration, Correlations, and Scatter Plots**
- **Performing Simple Linear Regression**
- **Producing Predicted Values**
- **Fitting a Multiple Linear Regression Model**
- **Automated Model Selection**
- **Estimating and Testing the Coefficients for the Selected Models**
- **Stepwise Regression**

Data Exploration, Correlations, and Scatter Plots

In exercise physiology, an objective measure of aerobic fitness is how efficiently the body can absorb and use oxygen (oxygen consumption). Subjects participated in a predetermined exercise run of 1.5 miles. Measurements of oxygen consumption as well as several other continuous measurements such as age, pulse, and weight were recorded. The researchers are interested in determining whether any of these other variables can help predict oxygen consumption. This data are found in Rawlings (1998) but certain values of **Maximum_Pulse** and **Run_Pulse** were changed for illustration. **Name**, **Gender**, and **Performance** were also modified for illustration.

The **fitness** data set contains the following variables:

Name	name of the member
Gender	gender of the member
Runtime	time to run 1.5 miles (in minutes)
Age	age of the member (in years)
Weight	weight of the member (in kilograms)
Oxygen_Consumption	a measure of the ability to use oxygen in the blood stream
Run_Pulse	pulse rate at the end of the run
Rest_Pulse	resting pulse rate
Maximum_Pulse	maximum pulse rate during the run
Performance	a measure of overall fitness

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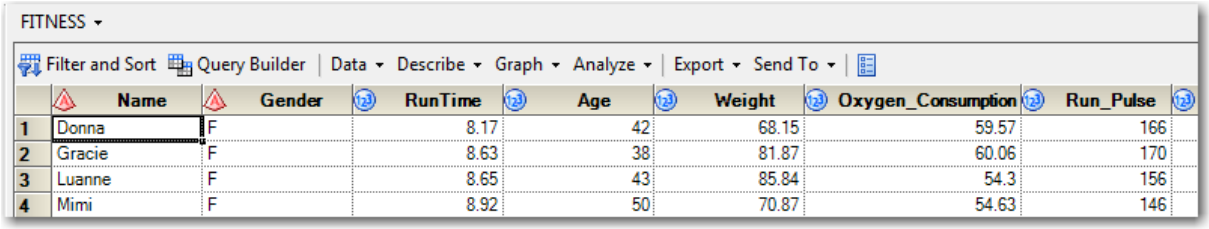
Examine the relationships between **Oxygen_Consumption** and the continuous predictor variables in the data set.

1. Create a new Process Flow and rename it **Correlation and Regression**.

2. Open the **Fitness** data set from the SASUSER library on the Local server.



FITNESS

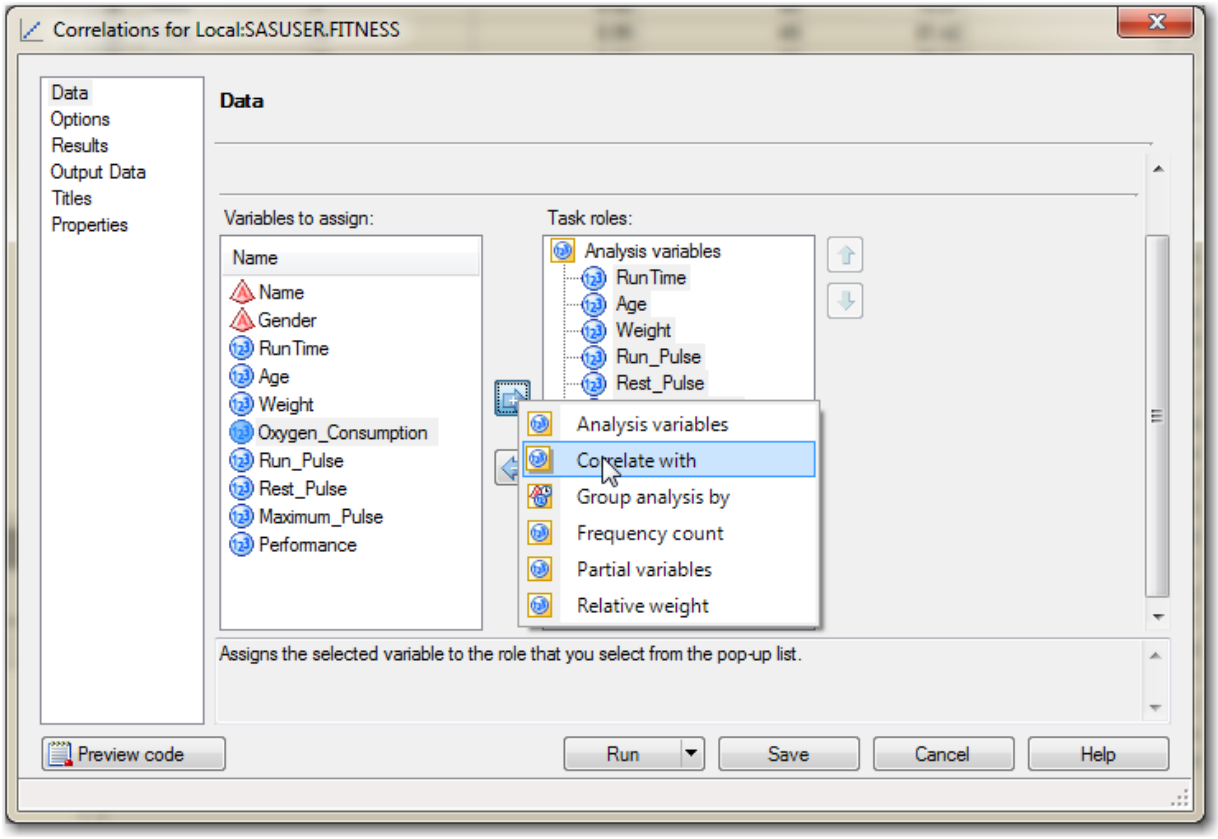
Filter and Sort | Query Builder | Data | Describe | Graph | Analyze | Export | Send To

	Name	Gender	RunTime	Age	Weight	Oxygen_Consumption	Run_Pulse
1	Donna	F	8.17	42	68.15	59.57	166
2	Gracie	F	8.63	38	81.87	60.06	170
3	Luanne	F	8.65	43	85.84	54.3	156
4	Mimi	F	8.92	50	70.87	54.63	146

3. Select **Analyze** ⇨ **Multivariate** ⇨ **Correlations**.

   You should also investigate the univariate statistics of continuous variables in the data set, just as you did in the earlier demos, using the Distribution Analysis task to explore distributions, measure central tendency and spread, and look for outliers.

4. With **Data** selected at the left, assign **Runtime**, **Age**, **Weight**, **Run_Pulse**, **Rest_Pulse**, **Maximum_Pulse**, and **Performance** to the task role of analysis variables and **Oxygen_Consumption** to the role of correlate with.



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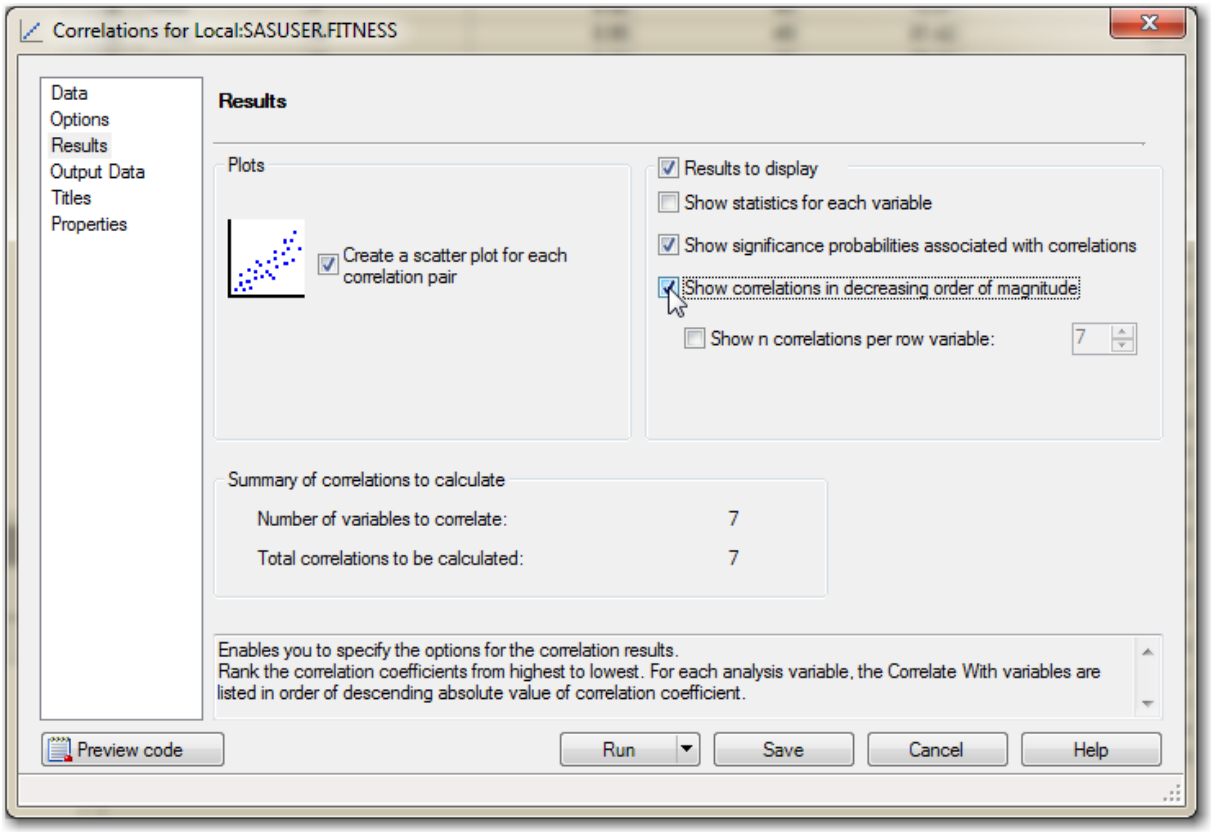
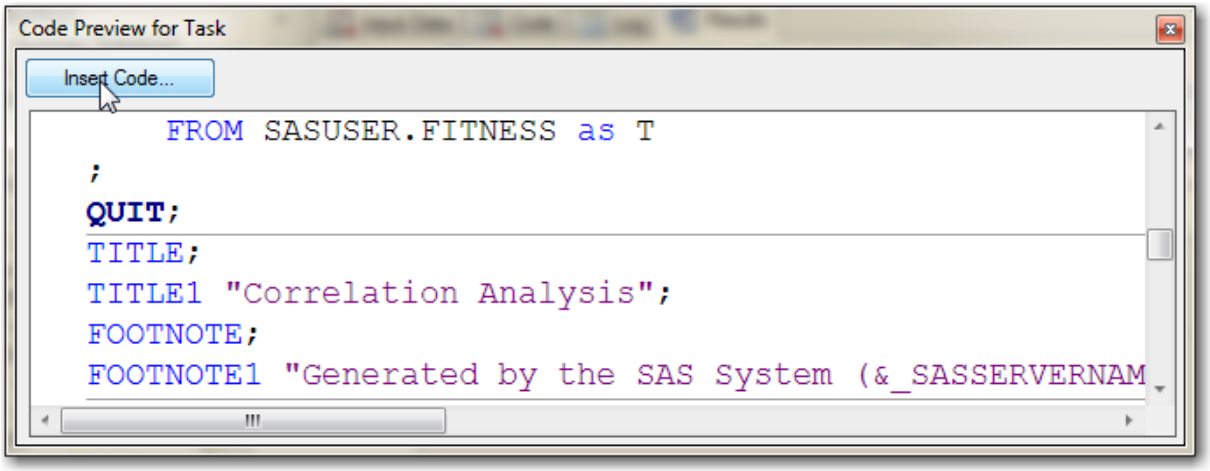
5. In **Results**, check the box for **Create a scatter plot for each correlation pair**. Also, check the box at the right for **Show correlations in decreasing order of magnitude** and uncheck the box for **Show statistics for each variable**.



By default, there will be a maximum of five scatter plots produced. If you want to allow for the creation of more than five scatter plots, then the code will need to be modified. The next steps explain how to do that.

6. Click **Preview code**.

7. Click **Insert Code** in the window that opens.



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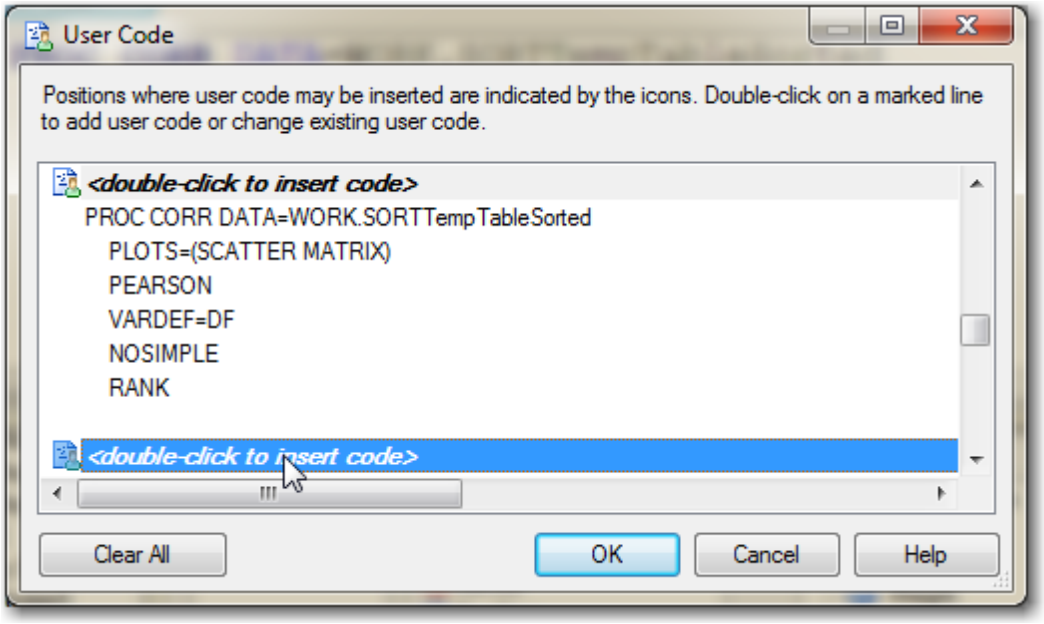
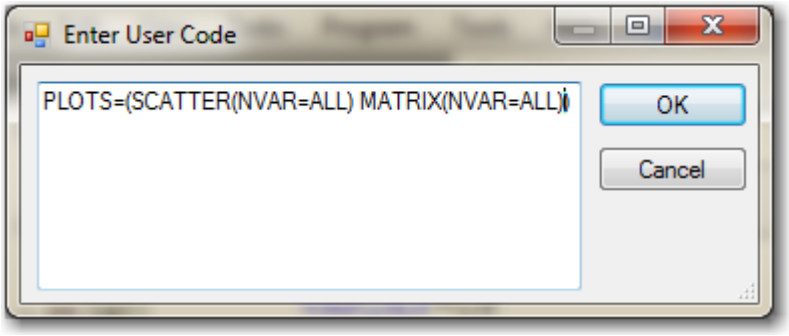
8. Double-click <***double-click to insert code***> below the word **RANK**.



9. Type the statement **PLOTS=(SCATTER(NVAR=ALL MATRIX(NVAR=ALL))** in the window and then click **OK** to close that window.



10. Click **OK** and then click ☒ to close out the next two windows.

11. Click **Run**.

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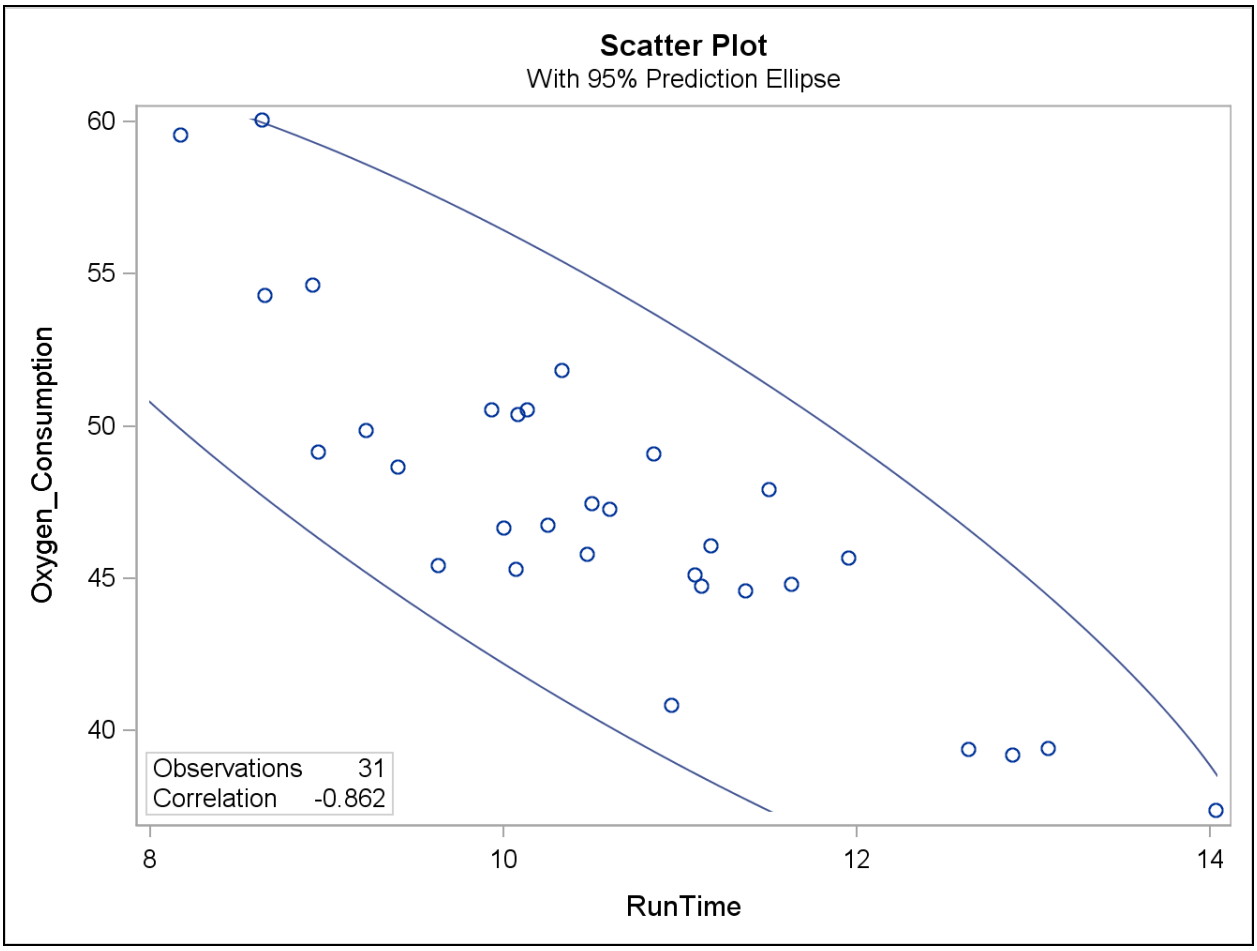
Output

1 With Variables:	Oxygen_Consumption
7 Variables:	RunTime Age Weight Run_Pulse Rest_Pulse Maximum_Pulse Performance

Pearson Correlation Coefficients, N = 31 Prob > |r| under H0: Rho=0							
Oxygen_Consumption	RunTime	Performance	Rest_Pulse	Run_Pulse	Age	Maximum_Pulse	Weight
	-0.86219	0.77890	-0.39935	-0.39808	-0.31162	-0.23677	-0.16289
	<.0001	<.0001	0.0260	0.0266	0.0879	0.1997	0.3813

The correlation coefficient between **Oxygen_Consumption** and **RunTime** is -0.86219. The *p*-value is small, which indicates that the population correlation coefficient (Rho) is likely different from 0. The second largest correlation coefficient, in absolute value, is **Performance**, at 0.77890.

Scatter plots associated with these correlations are shown below.



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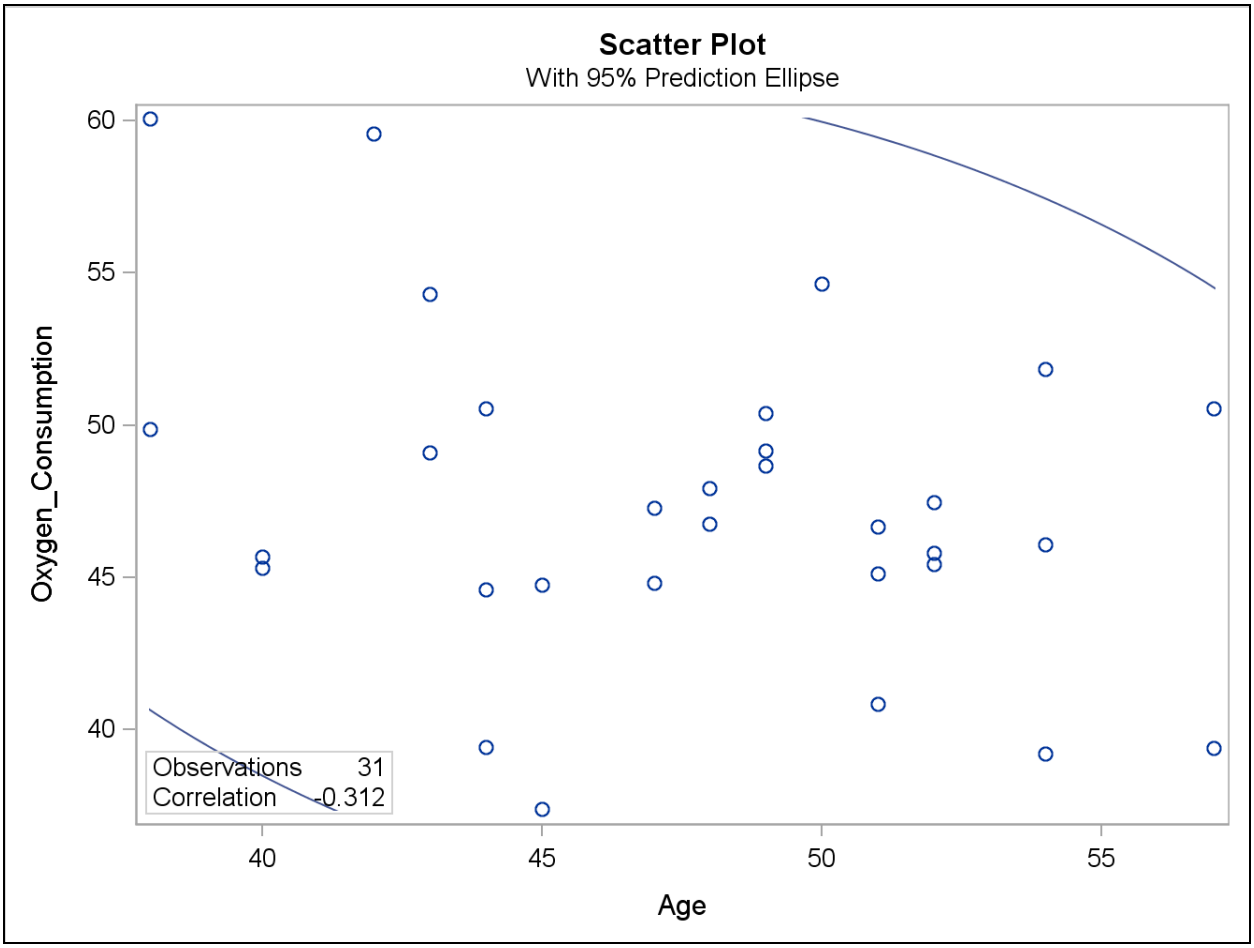

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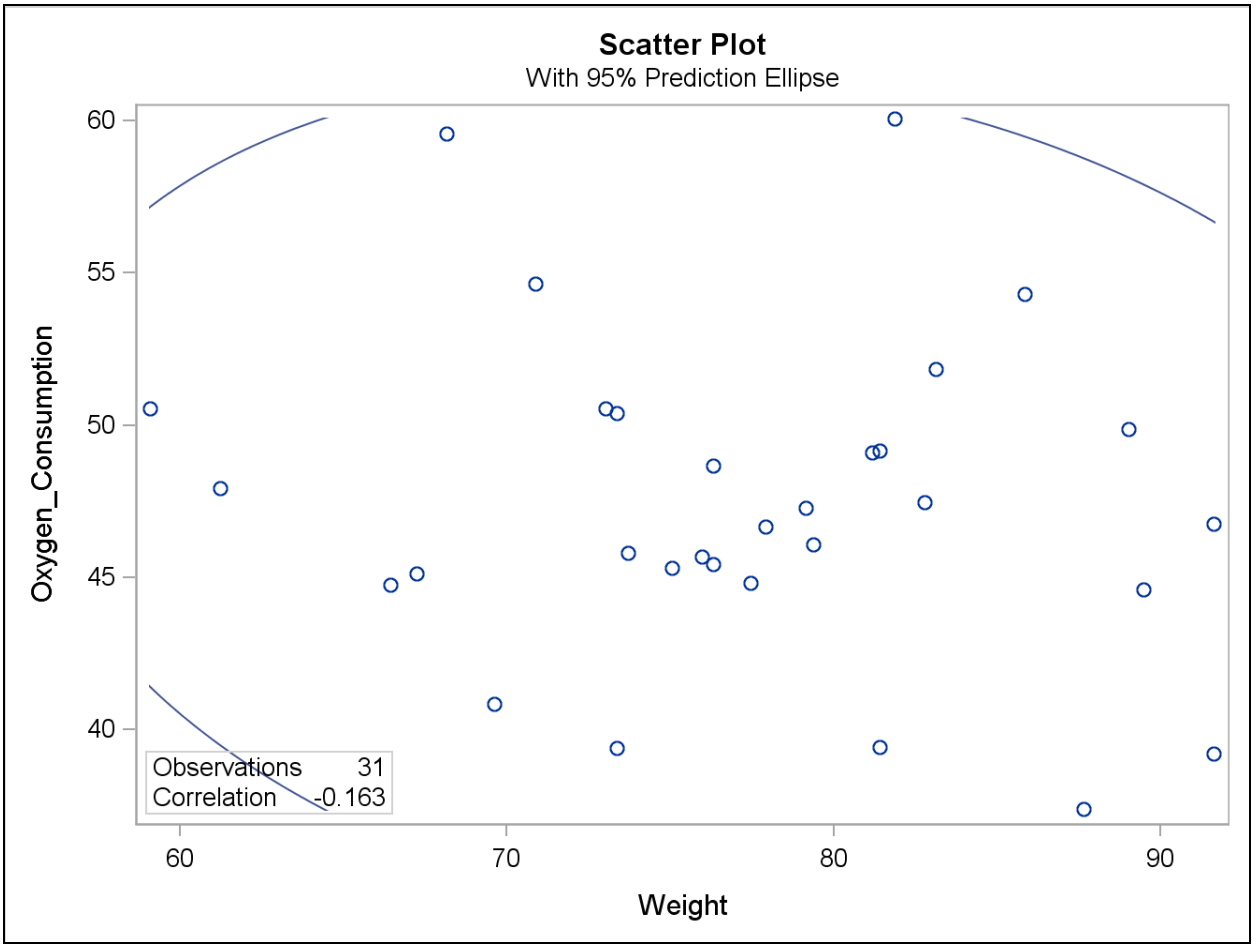

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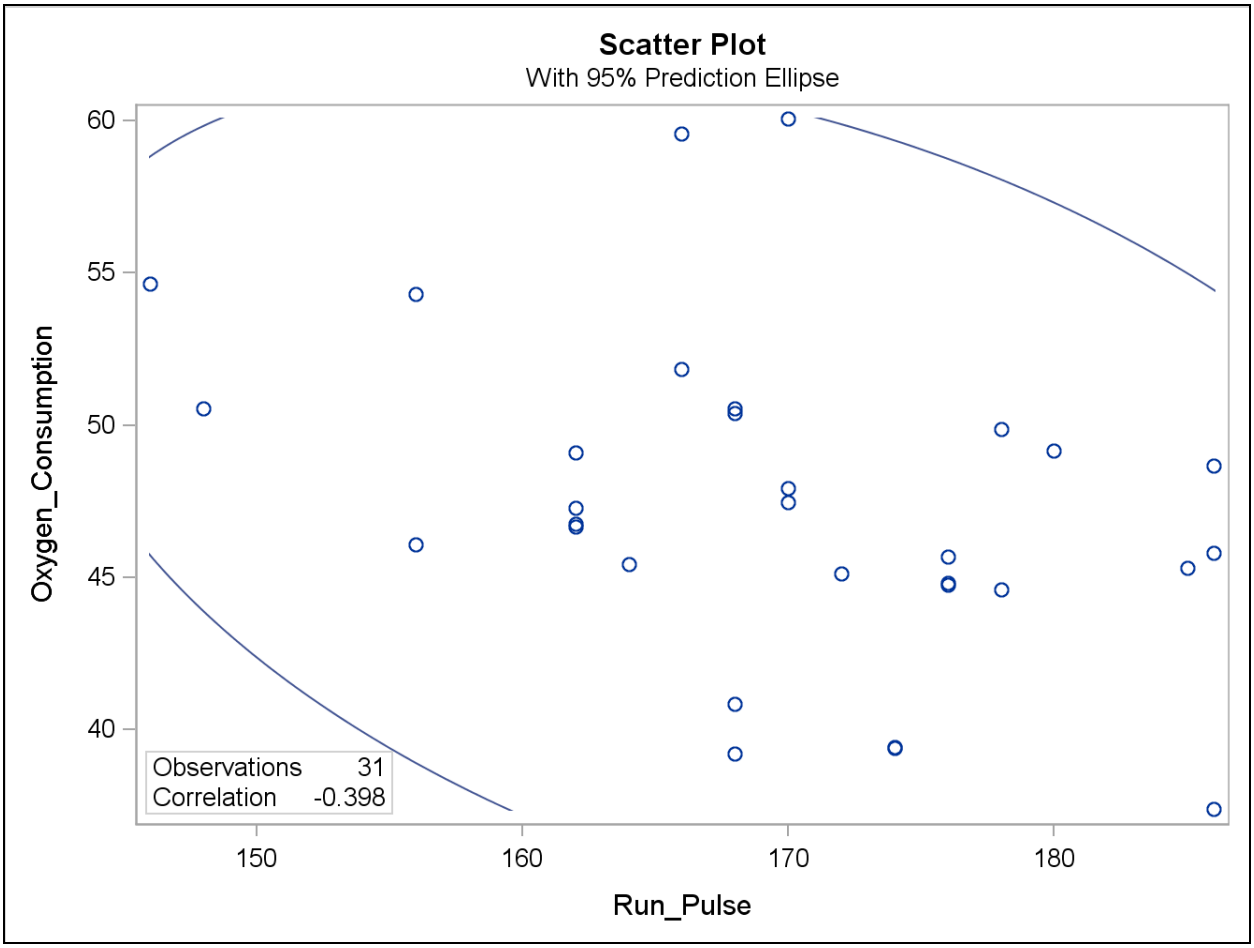

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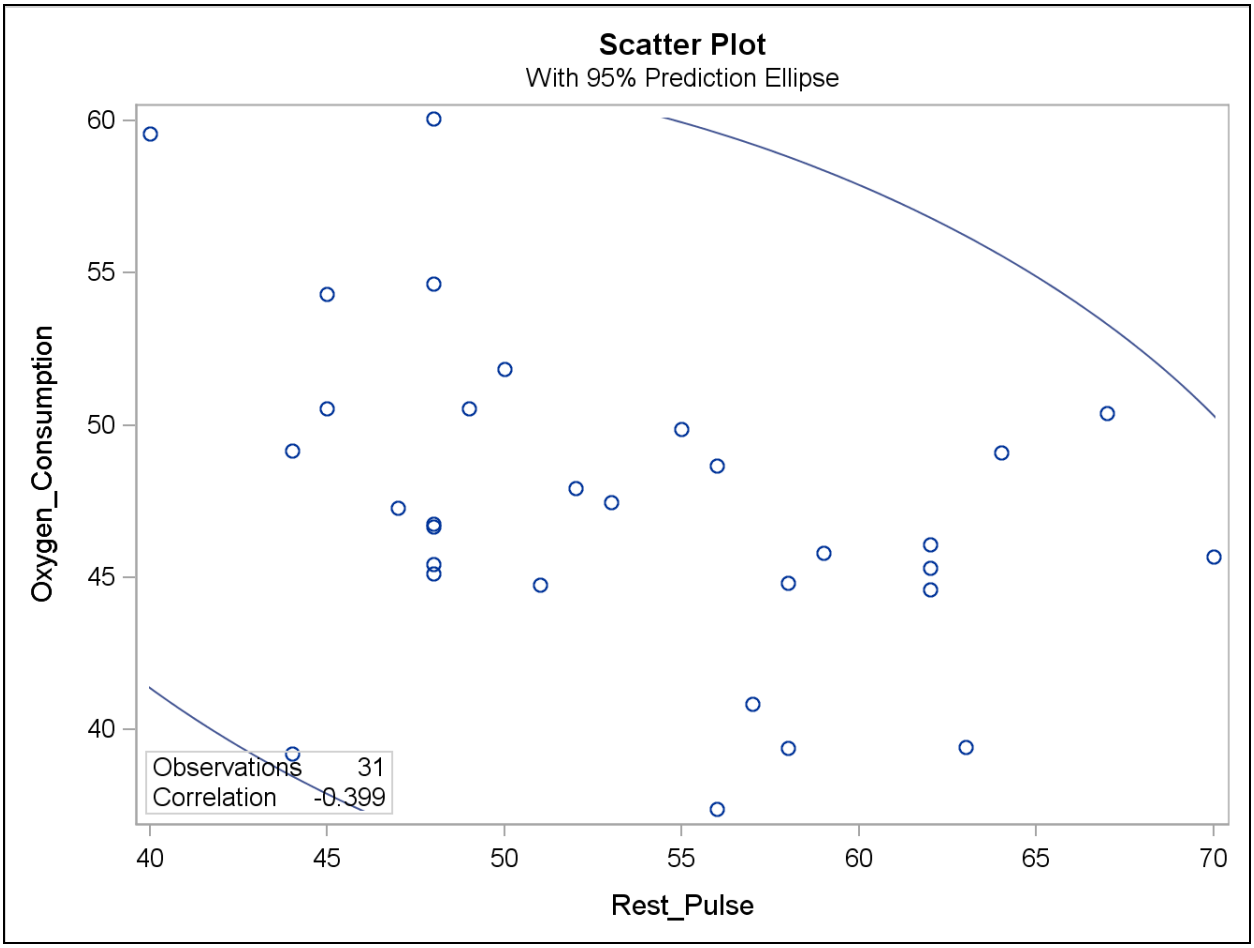

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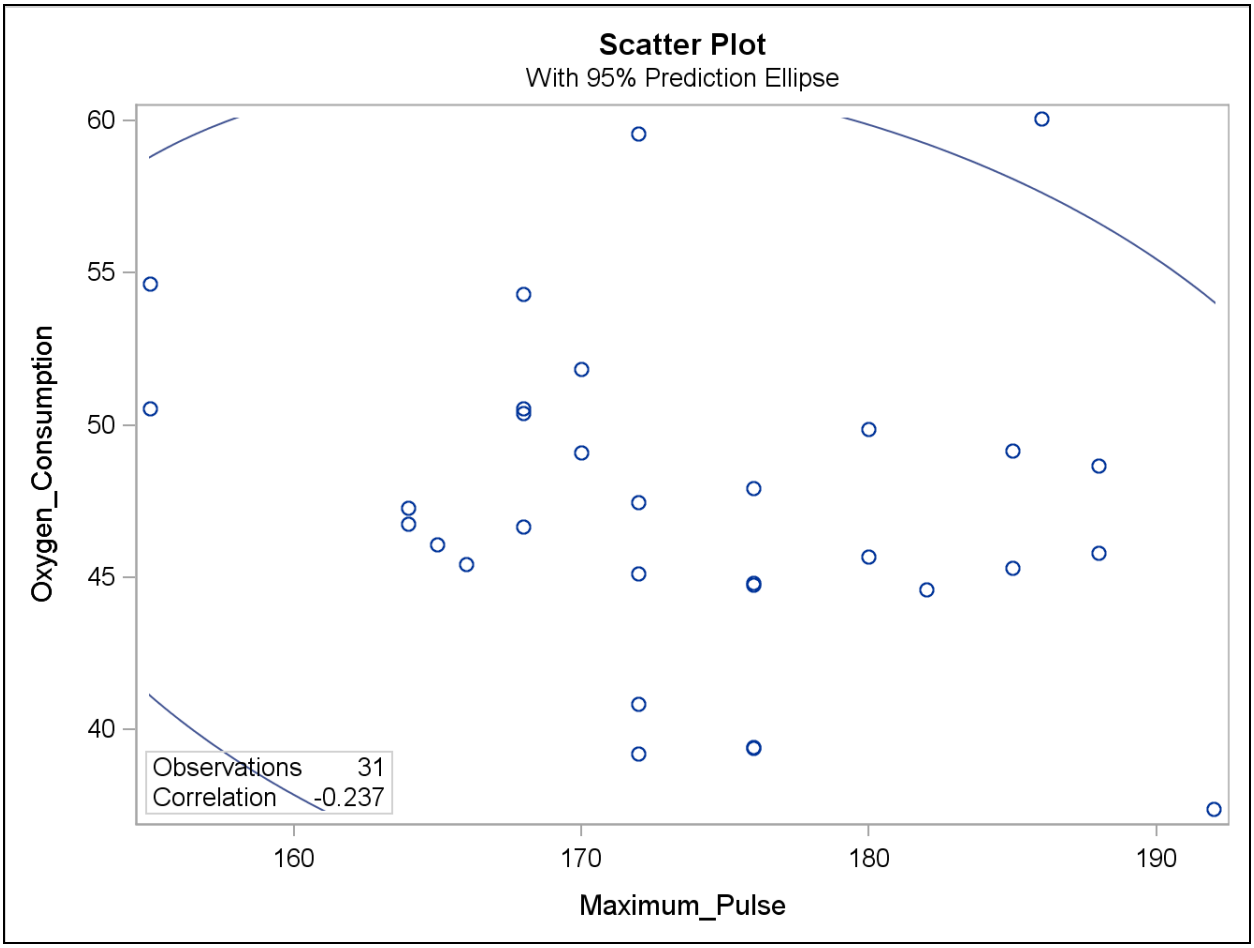

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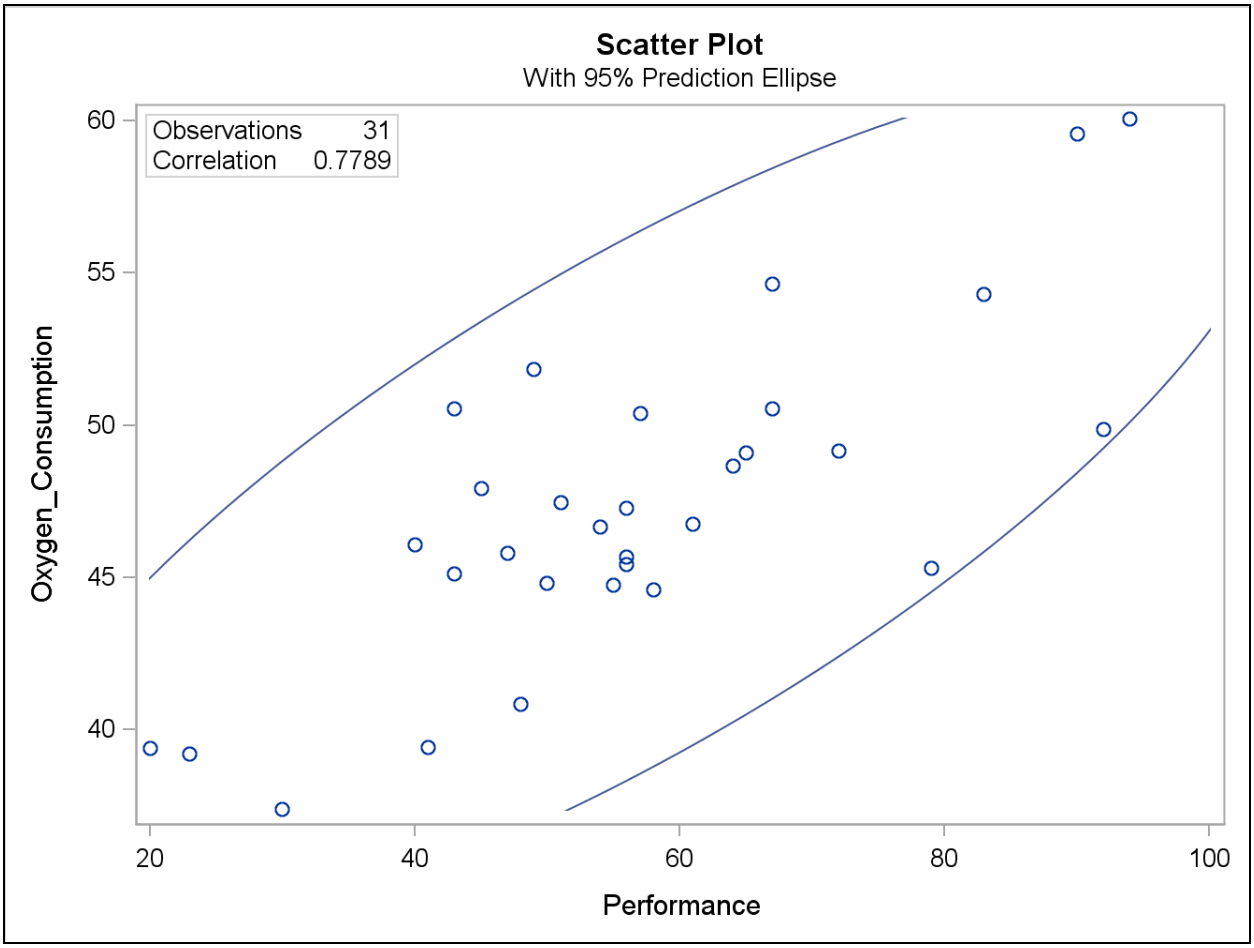
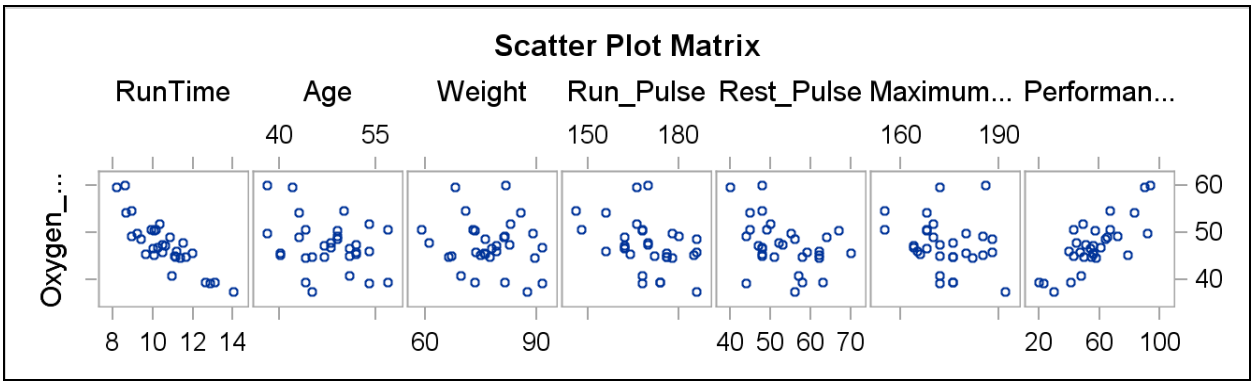


The correlation and scatter plot analyses indicate that several variables might be good predictors for **Oxygen_Consumption**.



The same scatter plots are displayed in the Scatter Plot Matrix.

When you prepare to conduct a regression analysis, it is always good practice to examine the correlations among the potential predictor variables. The Correlations task can be used to generate a matrix of correlation coefficients.

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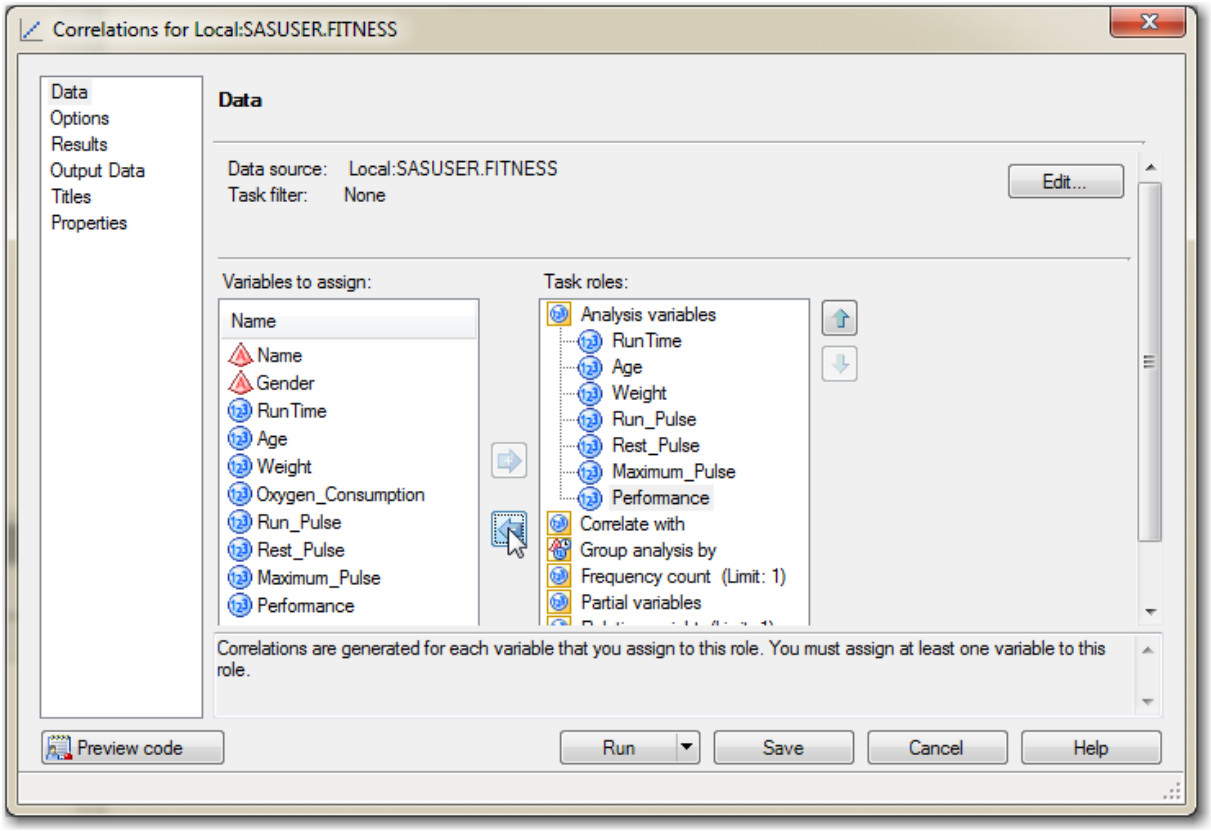
12. Reopen the previous task by right-clicking it and selecting **Modify**.
13. With **Data** selected at left, remove **Oxygen_Consumption** from the **Task roles** list by highlighting it and then clicking the button between the **Variables to assign** list and the **Task roles** list.



14. Click **Run**. Do not replace the results from the previous run.

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Output

7 Variables:	RunTime	Age	Weight	Run_Pulse	Rest_Pulse	Maximum_Pulse	Performance

Pearson Correlation Coefficients, N = 31 Prob > |r| under H0: Rho=0							
RunTime	RunTime 1.00000	Performance -0.82049 <.0001	Rest_Pulse 0.45038 0.0110	Run_Pulse 0.31365 0.0858	Maximum_Pulse 0.22610 0.2213	Age 0.19523 0.2926	Weight 0.14351 0.4412
Age	Age 1.00000	Performance -0.71257 <.0001	Maximum_Pulse -0.41490 0.0203	Run_Pulse -0.31607 0.0832	Weight -0.24050 0.1925	RunTime 0.19523 0.2926	Rest_Pulse -0.15087 0.4178
Weight	Weight 1.00000	Maximum_Pulse 0.24938 0.1761	Age -0.24050 0.1925	Run_Pulse 0.18152 0.3284	RunTime 0.14351 0.4412	Performance 0.08974 0.6312	Rest_Pulse 0.04397 0.8143
Run_Pulse	Run_Pulse 1.00000	Maximum_Pulse 0.92975 .0001	Rest_Pulse 0.35246 0.0518	Age -0.31607 0.0832	RunTime 0.31365 0.0858	Weight 0.18152 0.3284	Performance -0.02943 0.8751
Rest_Pulse	Rest_Pulse 1.00000	RunTime 0.45038 0.0110	Run_Pulse 0.35246 0.0518	Maximum_Pulse 0.30512 0.0951	Performance -0.22560 0.2224	Age -0.15087 0.4178	Weight 0.04397 0.8143
Maximum_Pulse	Maximum_Pulse 1.00000	Run_Pulse 0.92975 <.0001	Age -0.41490 0.0203	Rest_Pulse 0.30512 0.0951	Weight 0.24938 0.1761	RunTime 0.22610 0.2213	Performance 0.09002 0.6301
Performance	Performance 1.00000	RunTime -0.82049 <.0001	Age -0.71257 <.0001	Rest_Pulse -0.22560 0.2224	Maximum_Pulse 0.09002 0.6301	Weight 0.08974 0.6312	Run_Pulse -0.02943 0.8751

There are strong correlations between **Run_Pulse** and **Maximum_Pulse** (0.92975) and between **Runtime** and **Performance** (-0.82049). These associations are seen in more detail in the matrix of scatter plots.

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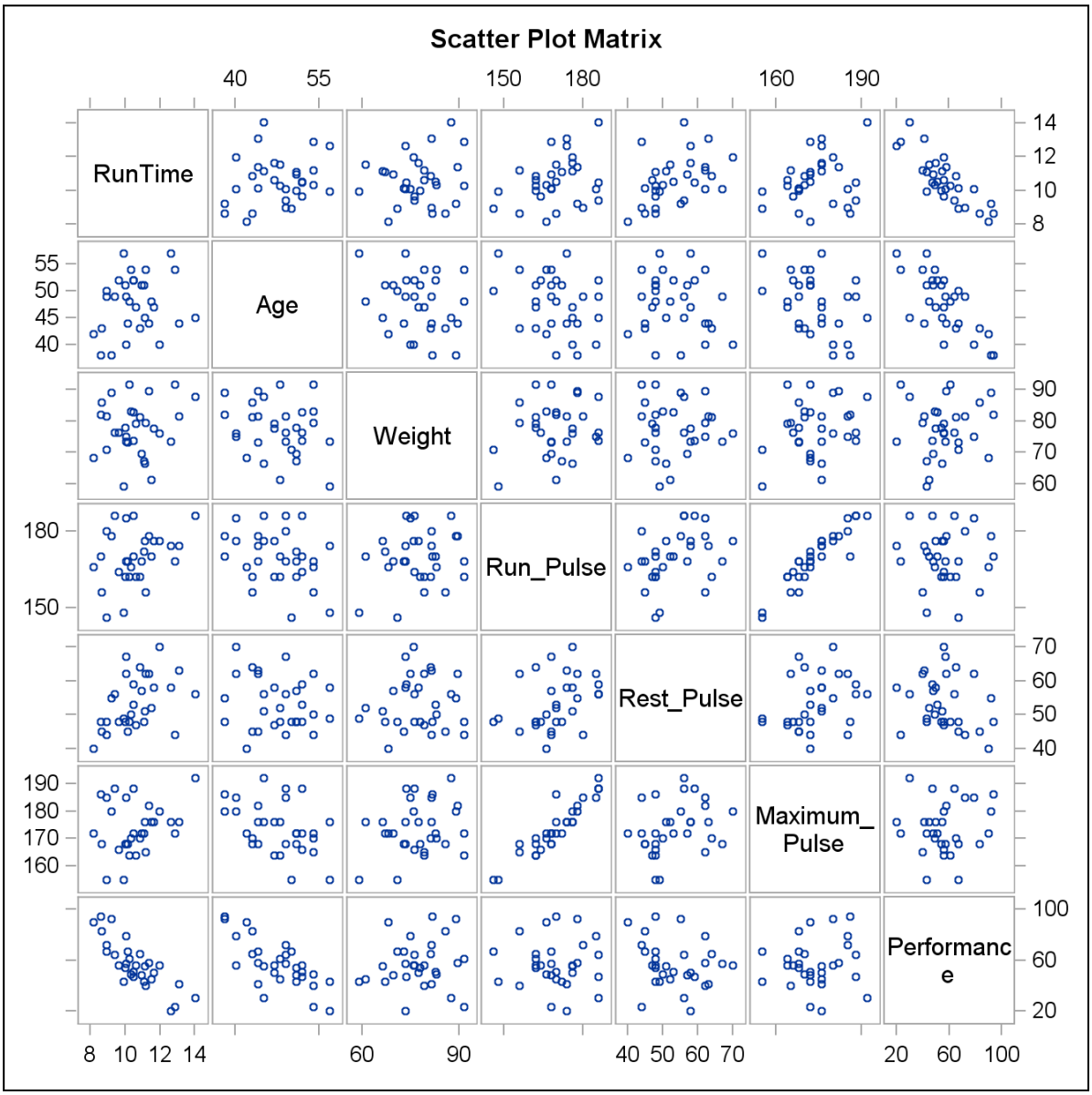
Skip to the Scatter PlotMatrix output:



The following correlation table was created from the matrix by choosing small *p*-values. The table is in descending order, based on the absolute value of the correlation. It provides a summary of the correlation analysis of the independent variables.

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Row Variable	Column Variable	Pearson's r	Prob > |r|
Run_Pulse	**Maximum_Pulse**	0.92975	<.0001
Runtime	**Performance**	-0.82049	<.0001
Performance	**Age**	-0.71257	<.0001
Runtime	**Rest_Pulse**	0.45038	0.0110
Age	**Maximum_Pulse**	-0.41490	0.0203
Run_Pulse	**Rest_Pulse**	0.35246	0.0518

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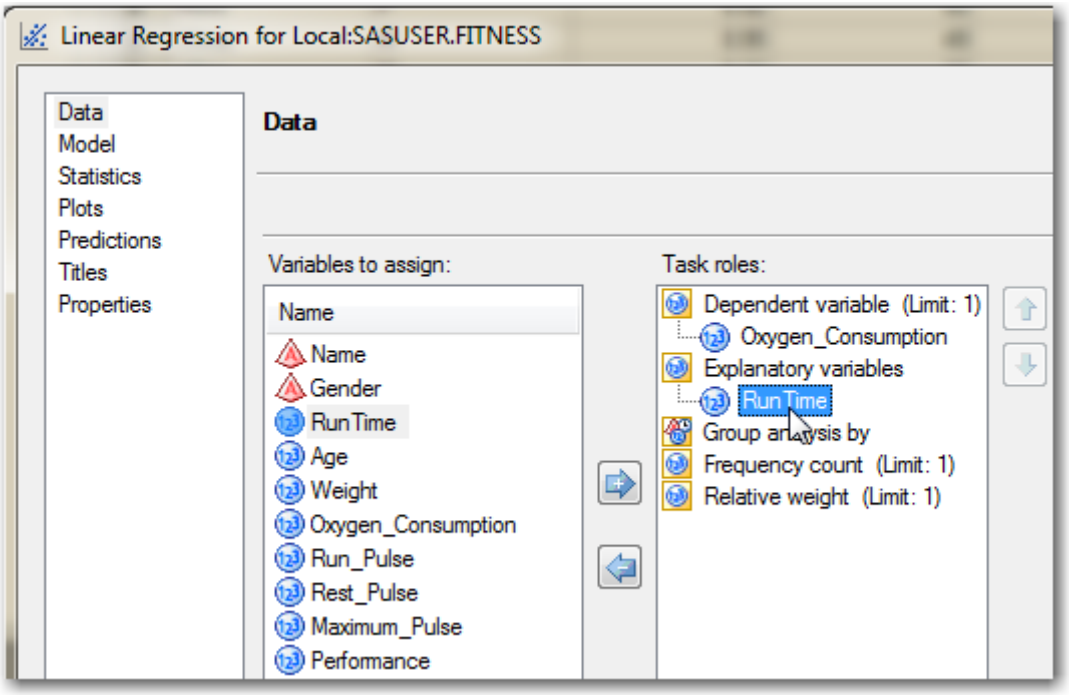
Performing Simple Linear Regression

Example: Because there is an apparent linear relationship between **Oxygen_Consumption** and **RunTime**, perform a simple linear regression analysis with **Oxygen_Consumption** as the response variable.

1. With the **Fitness** data set selected, select **Tasks** ⇨ **Regression** ⇨ **Linear Regression**.

2. Drag **Oxygen_Consumption** to the dependent variable task role and **RunTime** to the explanatory variables task role.



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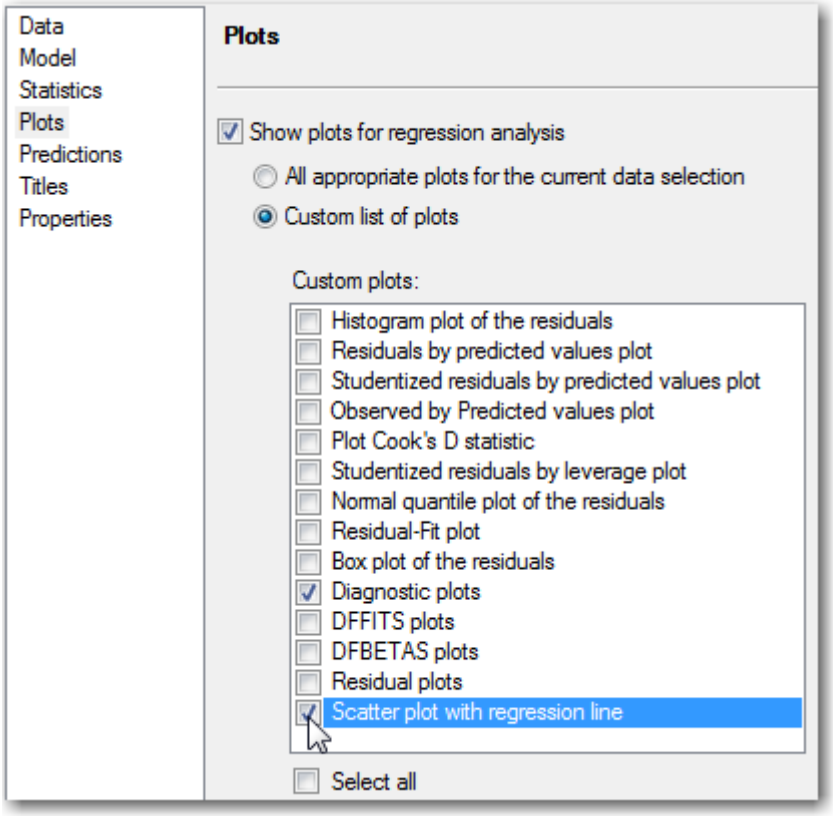
3. With **Plots** selected at the left, select **Custom list of plots** under **Show plots for regression analysis**. In the menu that appears check the box for **Scatter plot with regression line**.



4. Change the title, if desired.

5. Click **Run**.

Output

Number of Observations Read	31
Number of Observations Used	31

The Number of Observations Read and the Number of Observations Used are the same, which indicates that no missing values were detected for **Oxygen_Consumption** and **RunTime**.

Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	1	633.01458	633.01458	84.00	<.0001
Error	29	218.53997	7.53586		
Corrected Total	30	851.55455			

The Analysis of Variance (ANOVA) table provides an analysis of the variability observed in the data and the variability explained by the regression line.

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The ANOVA table for simple linear regression is divided into six columns.

Source labels the source of variability.

Model is the variability explained by your model (Between Group).

Error is the variability unexplained by your model (Within Group).

Corrected Total
is the total variability in the data (Total).

DF is the degrees of freedom associated with each source of variability.

Sum of Squares
is the amount of variability associated with each source of variability.

Mean Square
is the ratio of the sum of squares and the degrees of freedom. This value corresponds to the amount of variability associated with each degree of freedom for each source of variation.

F Value
is the ratio of the mean square for the model and the mean square for the error. This ratio compares the variability explained by the regression line to the variability unexplained by the regression line.

Pr > F
is the *p*-value associated with the *F* value.

The *F* value tests whether the slope of the predictor variable is equal to 0. The *p*-value is small (less than .05), so you have enough evidence at the .05 significance level to reject the null hypothesis. Thus, you can conclude that the simple linear regression model fits the data better than the baseline model. In other words, **RunTime** explains a significant amount of variability of **Oxygen_Consumption**.

The third part of the output provides summary measures of fit for the model.

Root MSE	2.74515	**R Square**	0.7434
Dependent Mean	47.37581	**Adj R Sq**	0.7345
Coeff Var	5.79442		

R-Square
The coefficient of determination is also referred to as the R^2 value. This value is

- between 0 and 1.
- the proportion of variability observed in the data explained by the regression line. In this example, the value is 0.7434, which means that the regression line explains 74% of the total variation in the response values.
- the square of the multiple correlation between Y and the Xs.

The R^2 is the squared value of the correlation that you saw earlier between **RunTime** and **Oxygen_Consumption** (0.86219). This is no coincidence. For simple regression, the R^2value is the square of the value of the bivariate Pearson correlation coefficient.

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Root MSE

The root mean square error is an estimate of the standard deviation of the response variable at each value of the predictor variable. It is the square root of the MSE.

Dependent Mean

The overall mean of the response variable - $\overline{Y}$.

Coeff Var

The coefficient of variation is the size of the standard deviation relative to the mean. The coefficient of variation is

- calculated as $\left(\frac{RootMSE}{\overline{Y}}\right) * 100$
- a unitless measure, so it can be used to compare data that have different units of measurement or different magnitudes of measurement.

Adj R-Sq

The adjusted R^2 is adjusted for the number of parameters in the model. This statistic is useful in multiple regression and is discussed in a later section.

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The Parameter Estimates table defines the model for your data.

Parameter Estimates					
Variable	**DF**	**Parameter Estimate**	**Standard Error**	**t Value**	**Pr > |t|**
Intercept	1	82.42494	3.85582	21.38	<.0001
RunTime	1	-3.31085	0.36124	-9.17	<.0001

DF represents the degrees of freedom associated with each term in the model.

Parameter Estimate
is the estimated value of the parameters associated with each term in the model.

Standard Error
is the standard error of each parameter estimate.

t Value is the *t* statistic, which is calculated by dividing the parameter estimates by their corresponding standard error estimates.

Pr > |t| is the *p*-value associated with the *t* statistic. It tests whether the parameter associated with each term in the model is different from 0. For this example, the slope for the predictor variable is statistically different from 0. Thus, you can conclude that the predictor variable explains a significant portion of variability in the response variable.

Because the estimate of β_o=82.42494 and β_1=–3.31085, the estimated regression equation is given by **Oxygen_Consumption**=82.42494–3.31085*(**RunTime**).

The model indicates that a one-unit greater value for **Runtime** is associated with a 3.31085 lesser value for **Oxygen_Consumption**. However, ***extrapolation of the model beyond the range of your predictor variables is inappropriate***. You cannot assume that the relationship maintains in areas that were not sampled from.

The parameter estimates table also shows that the intercept parameter is not equal to 0. However, the test for the intercept parameter only has practical significance when the range of values for the predictor variable includes 0. In this example, the test could not have practical significance because **RunTime**=0 (running at the speed of light) is not inside the range of observed values.

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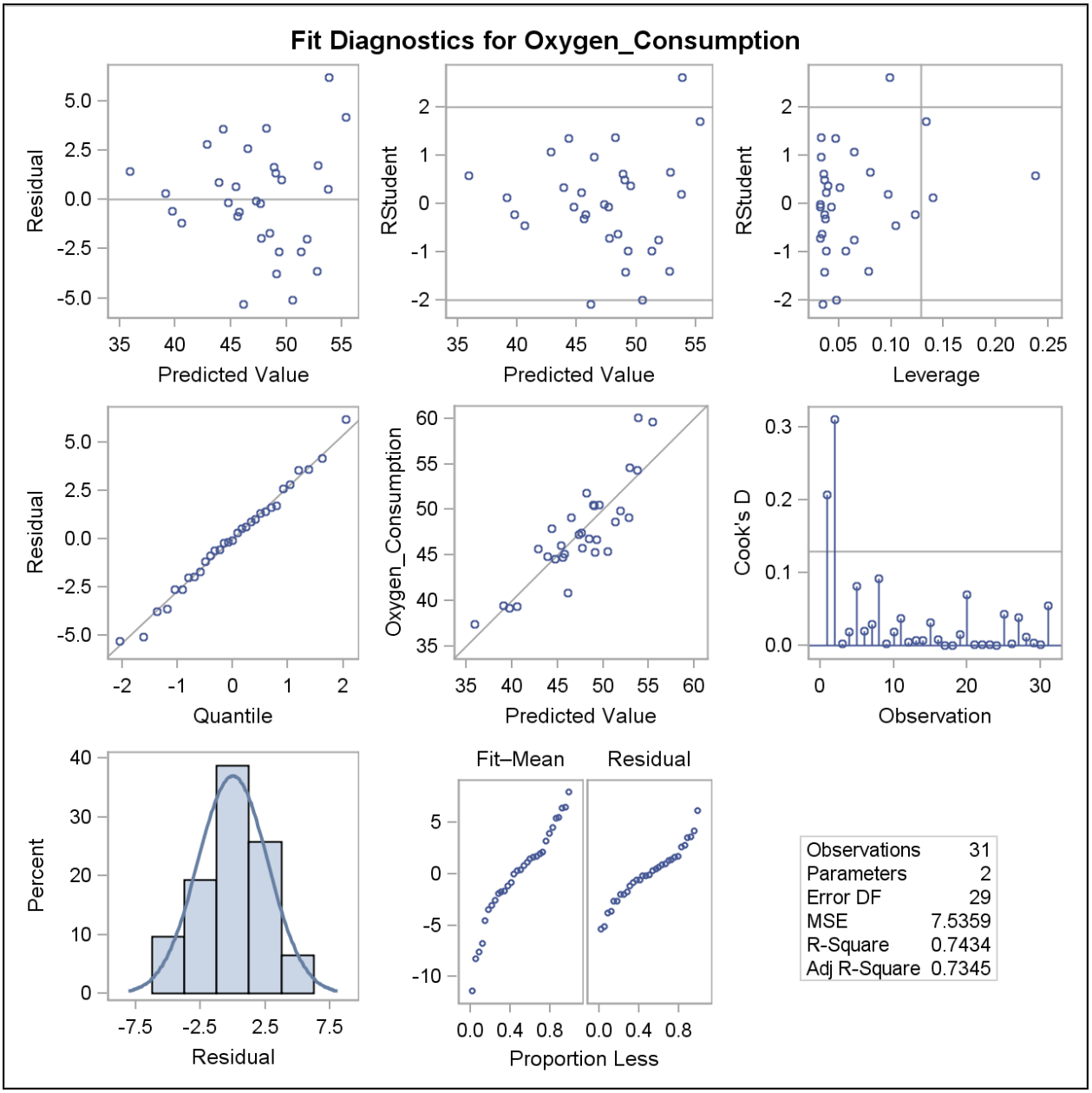

The diagnostics graph shows a variety of plots designed to help assess the data's fulfillment of statistical assumptions and show influential outliers.

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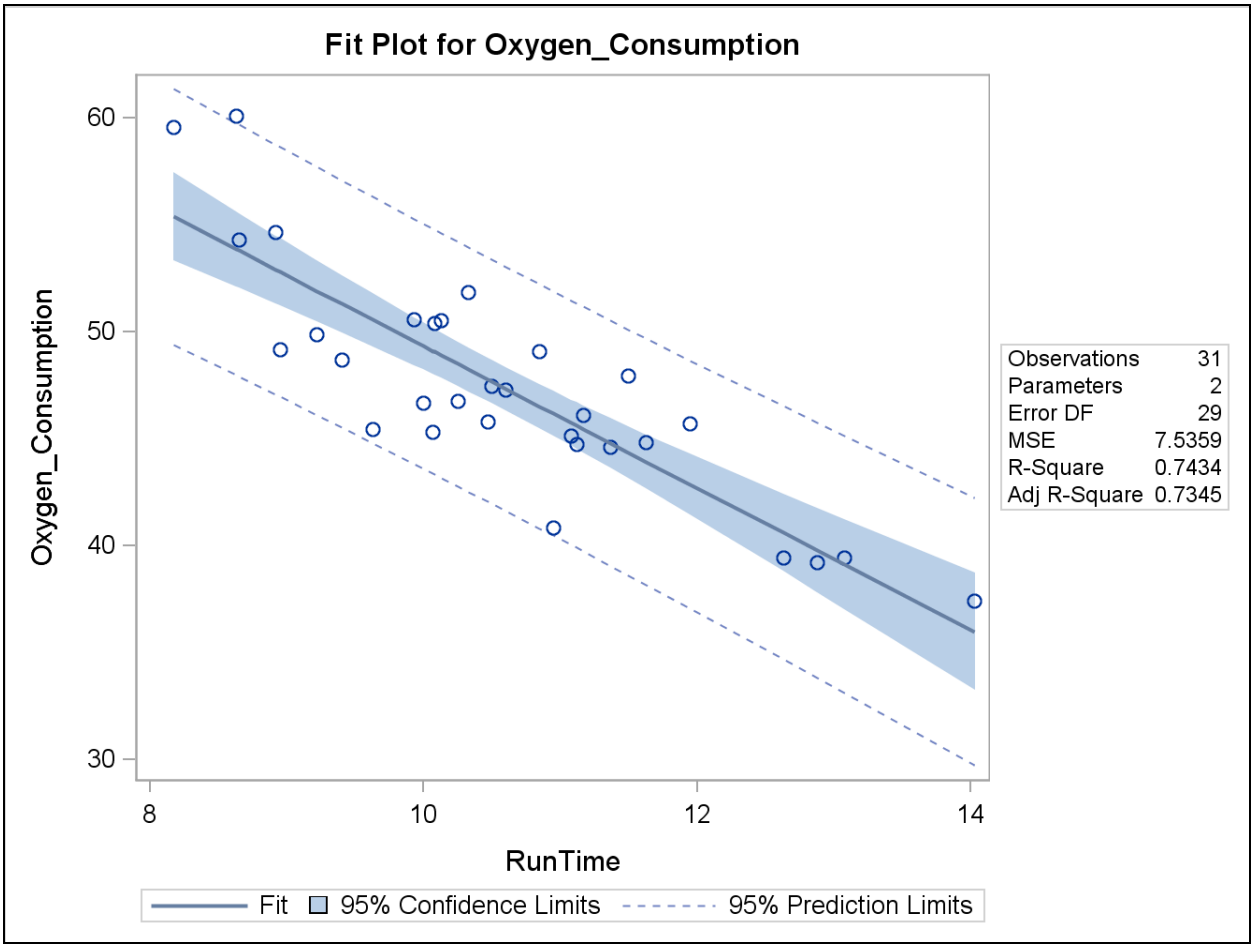

The Fit Plot produced by Enterprise Guide shows the predicted regression line superimposed over a scatter plot of the data.

To assess the level of precision around the mean estimates of **Oxygen_Consumption**, you can produce ***confidence intervals around the means***. This is represented in the shaded area in the plot.

- A 95% confidence interval for the mean says that you are 95% confident that your interval contains the population mean of Y for a particular X.
- Confidence intervals become wider as you move away from the mean of the independent variable. This reflects the fact that your estimates become more variable as you move away from the means of X and Y.

Suppose that the mean **Oxygen_Consumption** at a fixed value of **RunTime** is not the focus. If you are interested in establishing an inference on a future single observation, you need a ***prediction interval around the individual observations***. This is represented by the area between the broken lines in the plot.

- A 95% prediction interval is one that you are 95% confident contains a new observation.
- Prediction intervals are wider than confidence intervals because single observations have more variability than sample means.

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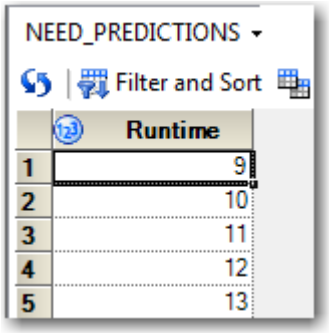
Producing Predicted Values

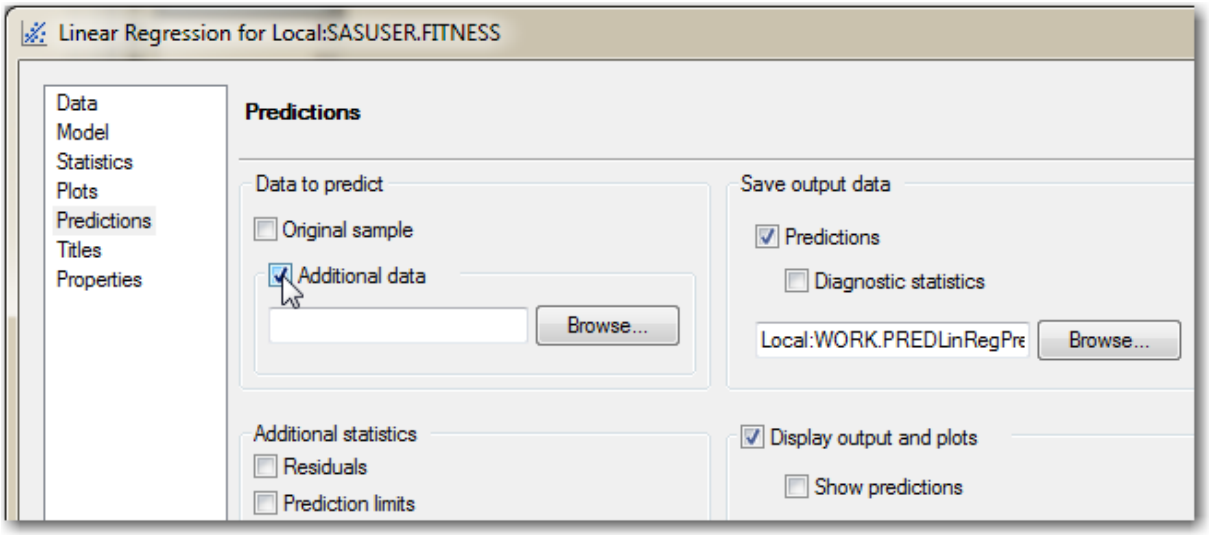
Example: Produce predicted values of **Oxygen_Consumption** when **RunTime** is 9, 10, 11, 12, or 13.

A data set called **Need_Predictions** contains 5 observations. The only variable in the data set is **RunTime**. It is displayed below:



1. Modify the previous Linear Regression task in the project.

2. With **Predictions** selected at the left, check the box for **Additional data**.



3. Click **Browse**.

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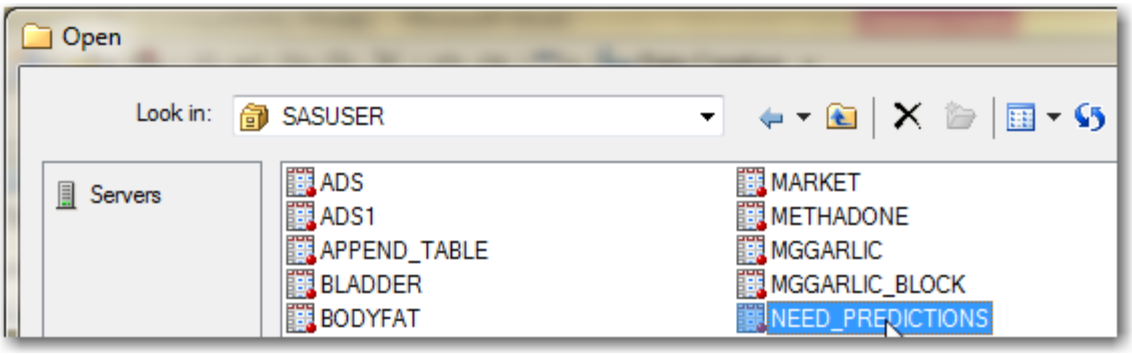
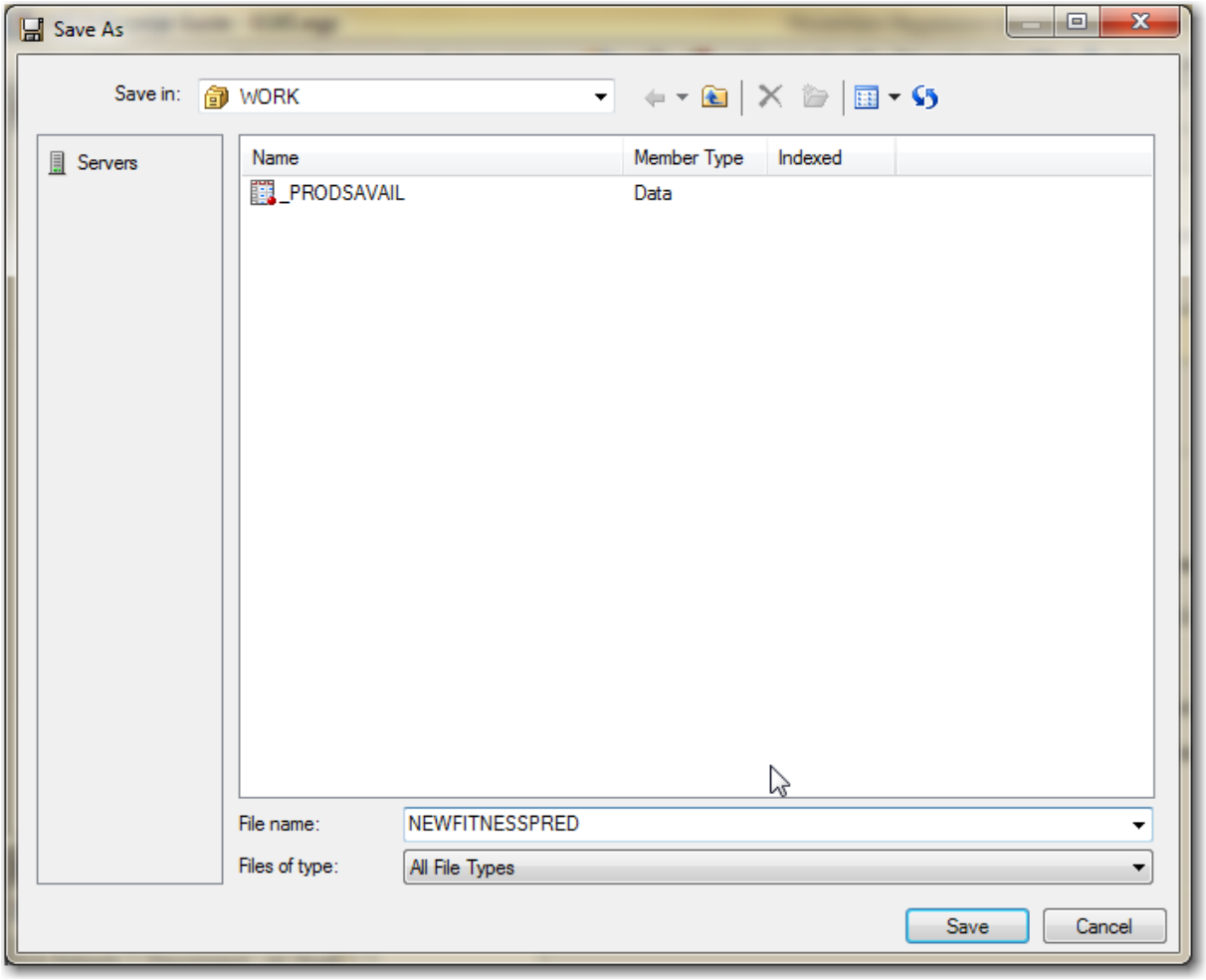
4. When the next window appears, select the **Need_Predictions** data set from the SASUSER library. Either double-click the name of the data set or highlight it and click **Open**.



5. Under **Save output data**, with **Predictions** checked, click **Browse**.

6. With that window open, type over the default filename with **NEWFITNESSPRED** (create it in the SASUSER library if you would like to make it permanent or in the WORK library if it will only be created for this session).



7. Click **Save** to close that window.

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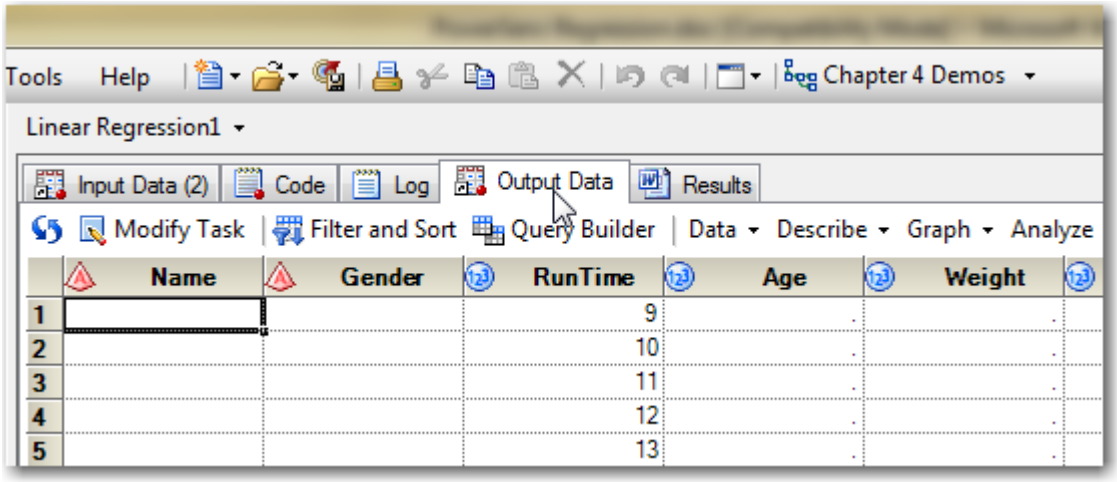
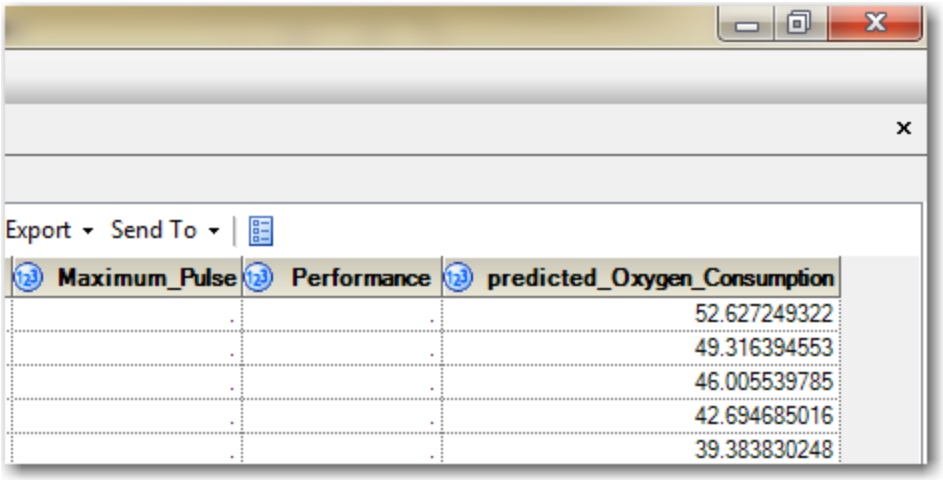
8. Click **Run**. Do not replace the results from the previous run.

9. In the workspace, you will now see a tab for the newly created data set, **NEWFITNESSPRED**. Click that tab to reveal the data.



10. Scroll all the way to the right to see the predictions column.



The predicted value for **Oxygen_Consumption** when **RunTime** is 9 is 52.627249322.

The new data set contains columns for all variables in the analysis data set, but the values of each record are set to missing for those variables that either were missing or did not exist in the scoring data set.

Choose only values within or near the range of the predictor variable when you are predicting new values for the response variable. For this example, the values of the variable **RunTime** range from 8.17 to 14.03 minutes. Therefore, it is unwise to predict the value of **Oxygen_Consumption** for a **RunTime** of 18. The reason is that the relationship between the predictor variable and the response variable might be different beyond the range of your data.

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Fitting a Multiple Linear Regression Model

Example: Perform a multiple linear regression analysis of **Oxygen_Consumption** on **Performance** and **Runtime**. Interpret the output for the two-variable model.

1. With the **Fitness** data set selected, select **Tasks** ⇨ **Regression** ⇨ **Linear Regression**.
2. Drag **Oxygen_Consumption** to the dependent variable task role and **RunTime** and **Performance** to the explanatory variables task role.
3. Uncheck the box for **Show plots for regression analysis** under **Plots**.
4. Change the title as desired and then click **Run**.

Output

Number of Observations Read	31
Number of Observations Used	31

Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	2	646.33101	323.16550	44.09	<.0001
Error	28	205.22355	7.32941		
Corrected Total	30	851.55455			

Model DF
is 2, the number of parameters minus 1.

Error DF
is 28, the total numbers of observations (31) minus the number of parameters in the model (3).

Corrected Total DF
is 30, the number of observations minus 1.

Model Sum of Squares
is the total variation in the Y explained by the model.

Error Sum of Squares
is the variation in the Y ***not*** explained by the model.

Corrected Total Sum of Squares is the total variation in the Y.

Model Mean Square
is the Model Sum of Squares divided by the Model DF, which is also known as model variance.

Mean Square Error
is the Error Sum of Squares divided by the Error DF and is an estimate of σ^2, the variance of the

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random error term, which is also known as error variance.

F Value
is the (Mean Square Model)/(Mean Square Error).

Pr > F is small. Therefore, you reject H_0: $\beta_1=\beta_2=0$ and conclude that at least one $\beta_i \neq 0$.

Root MSE	2.70729	**R-Square**	0.7590
Dependent Mean	47.37581	**Adj R-Sq**	0.7418
Coeff Var	5.71450		

The R^2 for this model, 0.7590, is only slightly larger than the R^2 for the model in which **RunTime** is the only predictor variable, 0.7434.

The adjusted R^2 for this model is 0.7418, slightly higher than the adjusted R^2 of 0.7345 for the **RunTime** only model. This suggests, although mildly, that adding **Performance** does improve the model predicting **Oxygen_Consumption**.

Parameter Estimates					
Variable	**DF**	**Parameter Estimate**	**Standard Error**	**t Value**	**Pr > |t|**
Intercept	1	71.52626	8.93520	8.00	<.0001
RunTime	1	-2.62163	0.62320	-4.21	0.0002
Performance	1	0.06360	0.04718	1.35	0.1885

Using the estimates for β_0, β_1, and β_2 above, this model can be written as

Oxygen_Consumption=71.56260.06360***Performance**–2.62163***Runtime**

The *p*-value for **Performance** is large, which suggests that the slope is not significantly different from 0. The correlation that you saw between **Performance** and **Oxygen_Consumption** was large and statistically significant (r=.77890, p<.0001). The test for $\beta_i=0$ is conditioned on the other terms in the model. That is the reason that neither **Performance** nor **RunTime** have the same *p*-values (or parameter estimates) when used alone as when used in a model that includes both. The test for $\beta_1=0$ (for **Performance**) is conditional on (or adjusted for) X_2 (**Runtime**). Similarly, the test for $\beta_2=0$ is conditional on X_1 (**Performance**).

The significance level of the test does ***not*** depend on the order in which you list the explanatory variables, but it does depend on the combination of explanatory variables present in the model.

In a later section, you look at the difficulties involved with analyzing and selecting the best models due to the relationships among predictor variables.

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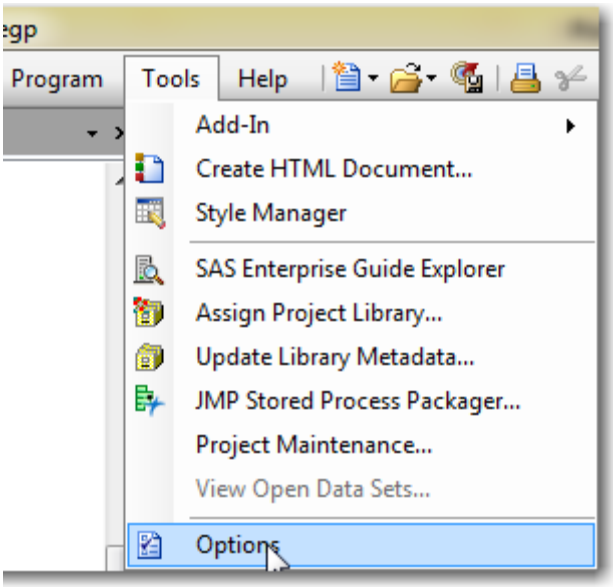
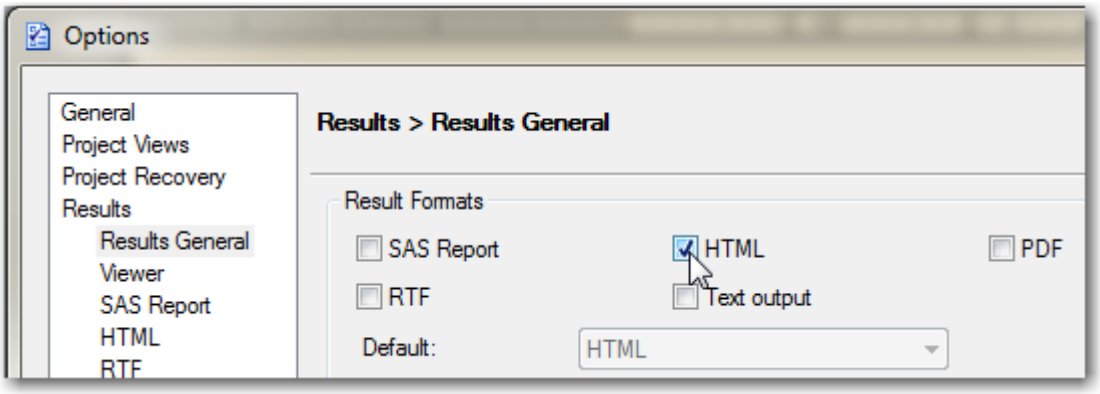
Automated Model Selection

Example: Invoke the Linear Regression Task to produce a regression of **Oxygen_Consumption** on all the other numeric variables in the **Fitness** data set.

1. Select **Tools** ⇨ **Options**.



2. In the window that opens, select **Results General** under **Results** at the left and then check the box for **HTML** if it is not already selected.



3. Click **OK**.

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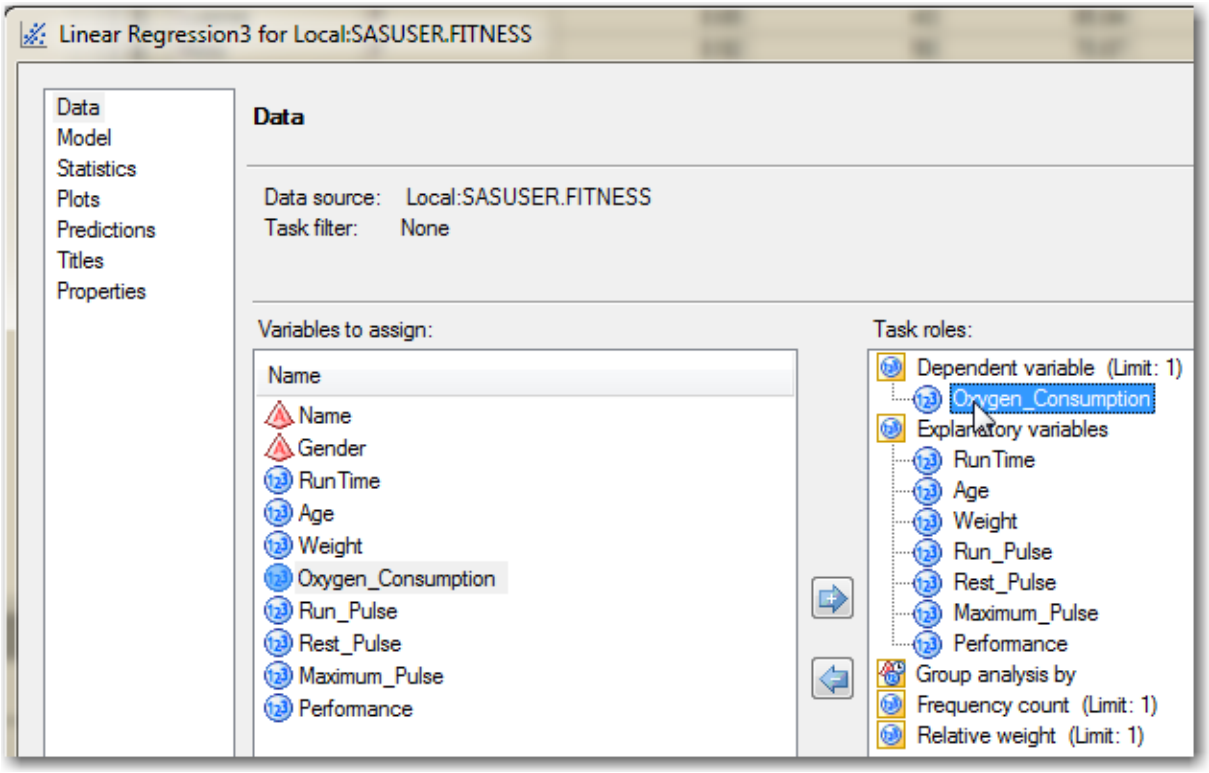
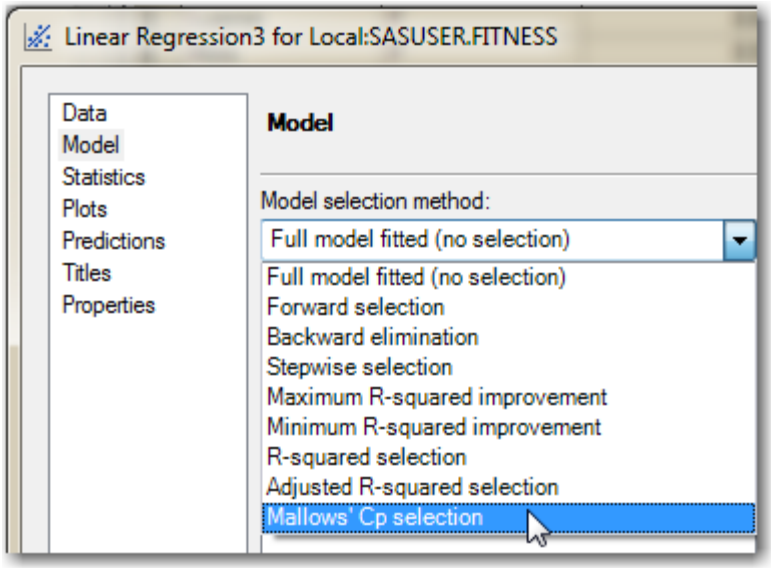
Now you are ready to run the Linear Regression task.

4. With the **Fitness** data set selected, select **Tasks** ⇨ **Regression** ⇨ **Linear Regression**.

5. Drag **Oxygen_Consumption** to the dependent variable task role and all other numeric variables to the explanatory variables task role.



6. With **Model** selected at the left, find the pull-down menu for **Model selection method** and click to find **Mallows' Cp selection** at the bottom.



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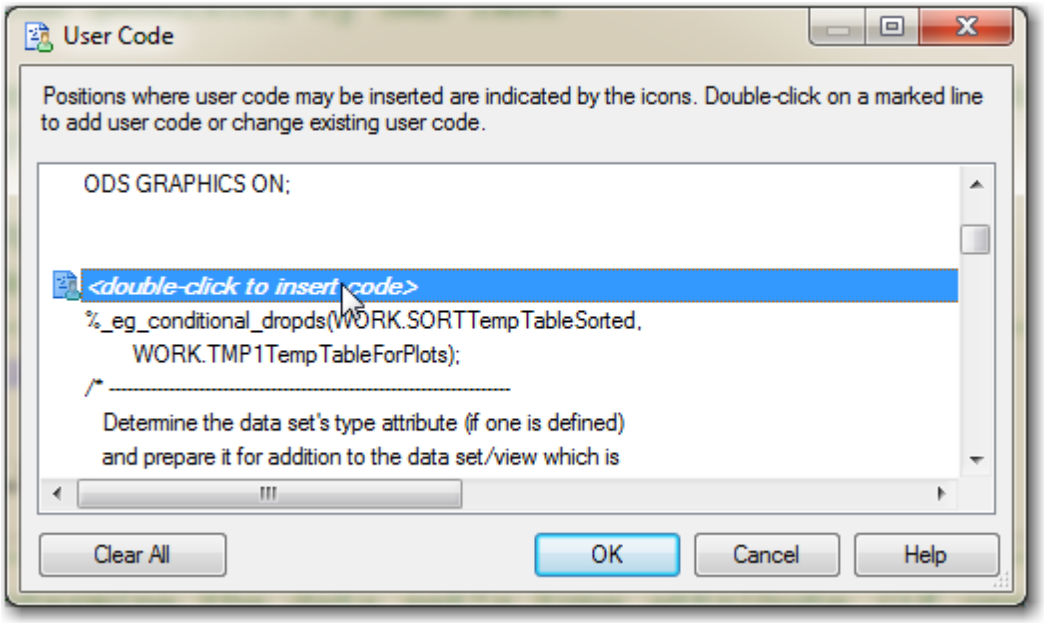
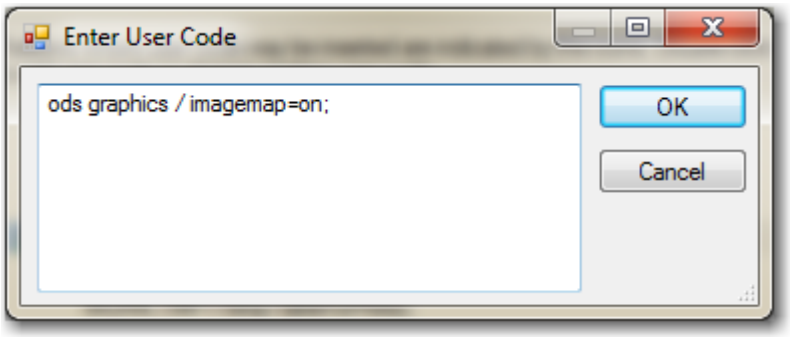
7. Click **Preview code**.

8. Click **Insert Code**.

9. Double-click ***<double-click to insert code>*** under the ODS GRAPHICS ON; statement.



10. Type **ODS GRAPHICS / IMAGEMAP=ON;** in the Enter User Code window.



11. Click **OK** and then click **OK** to close the last two windows opened. Click in the Code Preview for Task window.

12. Click **Run**.

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Partial HTML Output

Number of Observations Read	31
Number of Observations Used	31

Model Index	Number in Model	C(p)	R-Square	Variables in Model
1	4	4.0004	0.8355	RunTime Age Run_Pulse Maximum_Pulse
2	5	4.2598	0.8469	RunTime Age Weight Run_Pulse Maximum_Pulse
3	5	4.7158	0.8439	RunTime Weight Run_Pulse Maximum_Pulse Performance
4	5	4.7168	0.8439	RunTime Age Run_Pulse Maximum_Pulse Performance
5	4	4.9567	0.8292	RunTime Run_Pulse Maximum_Pulse Performance
6	3	5.8570	0.8101	RunTime Run_Pulse Maximum_Pulse
7	3	5.9367	0.8096	RunTime Age Run_Pulse
8	5	5.9783	0.8356	RunTime Age Run_Pulse Rest_Pulse Maximum_Pulse
9	5	5.9856	0.8356	Age Weight Run_Pulse Maximum_Pulse Performance
10	6	6.0492	0.8483	RunTime Age Weight Run_Pulse Maximum_Pulse Performance
11	6	6.1758	0.8475	RunTime Age Weight Run_Pulse Rest_Pulse Maximum_Pulse
12	6	6.6171	0.8446	RunTime Weight Run_Pulse Rest_Pulse Maximum_Pulse Performance

There are many models to compare. It would be unwieldy to try to determine the best model by viewing the output tables. Therefore, it is advisable to look at the ODS plots.

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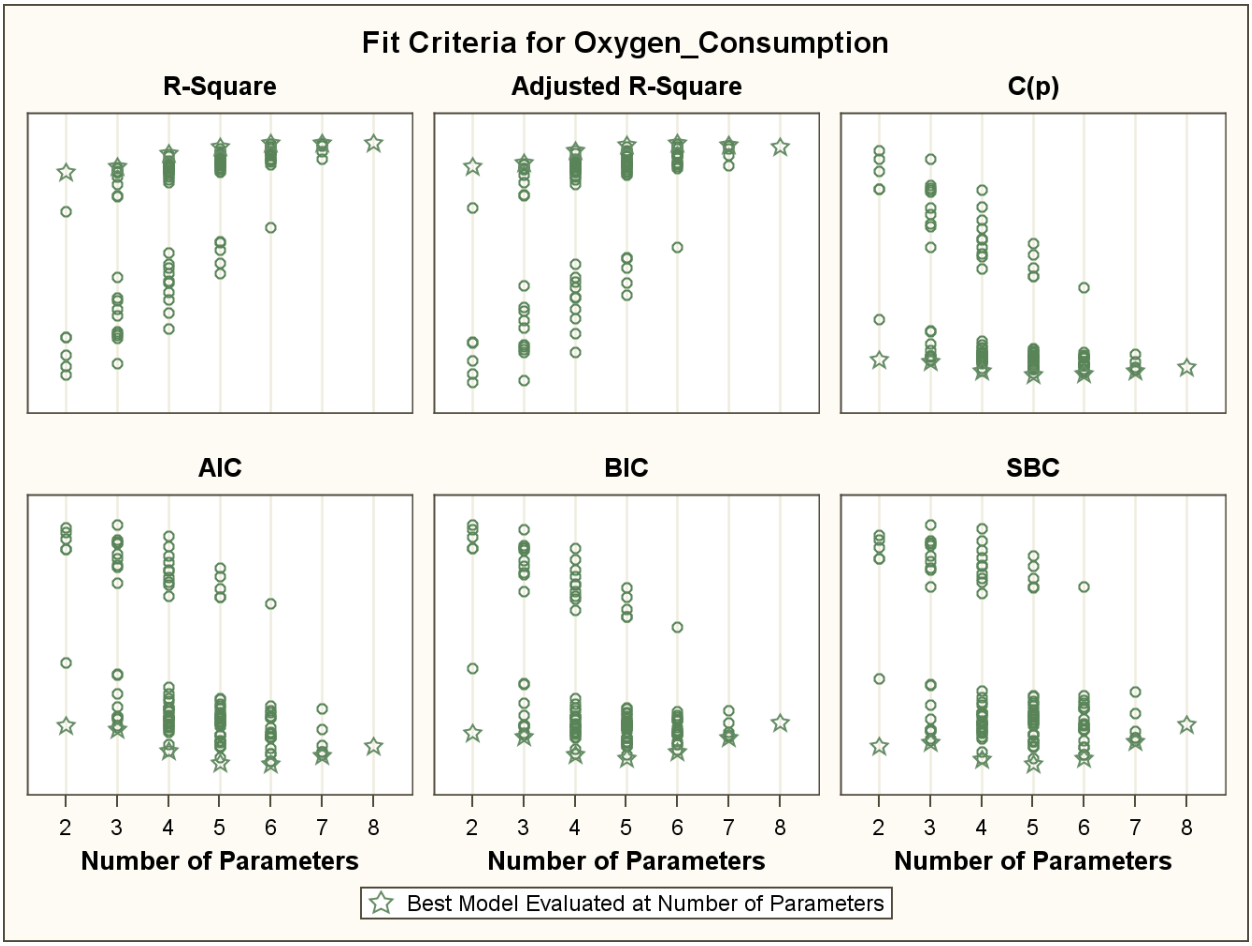

The first plot is a panel plot of several plots assessing each of the 127 possible subset models. Three of them will be further described below.

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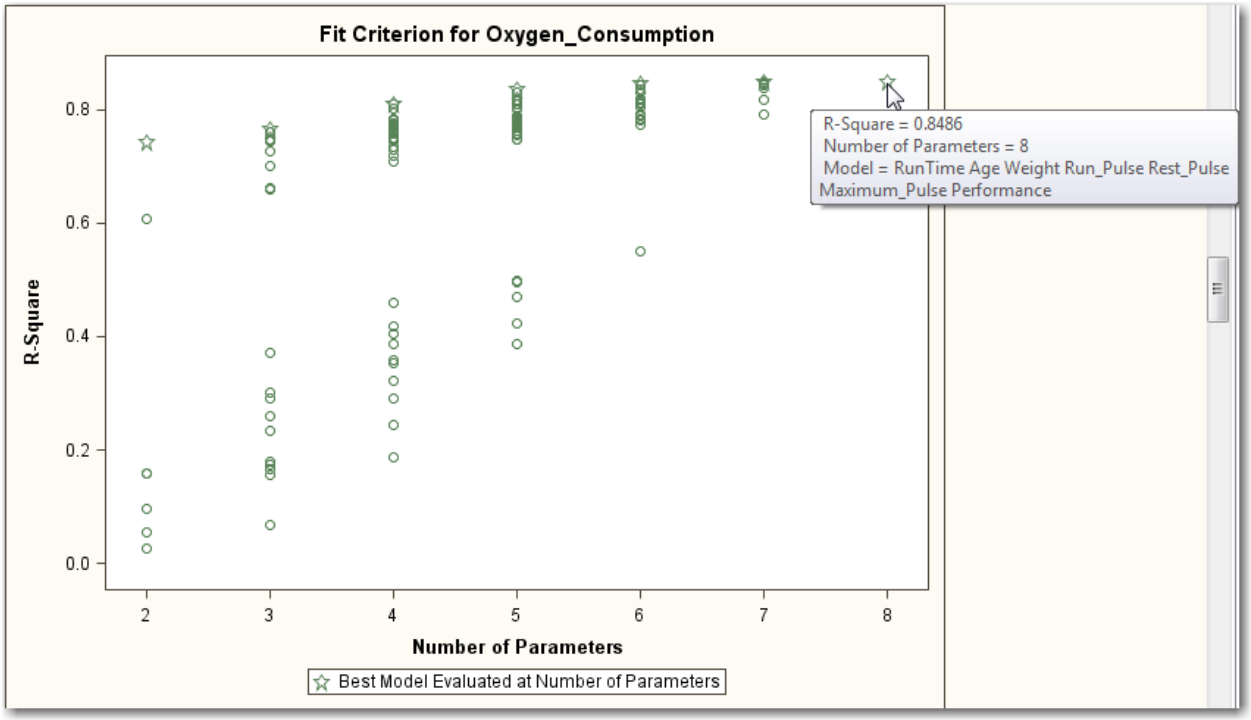
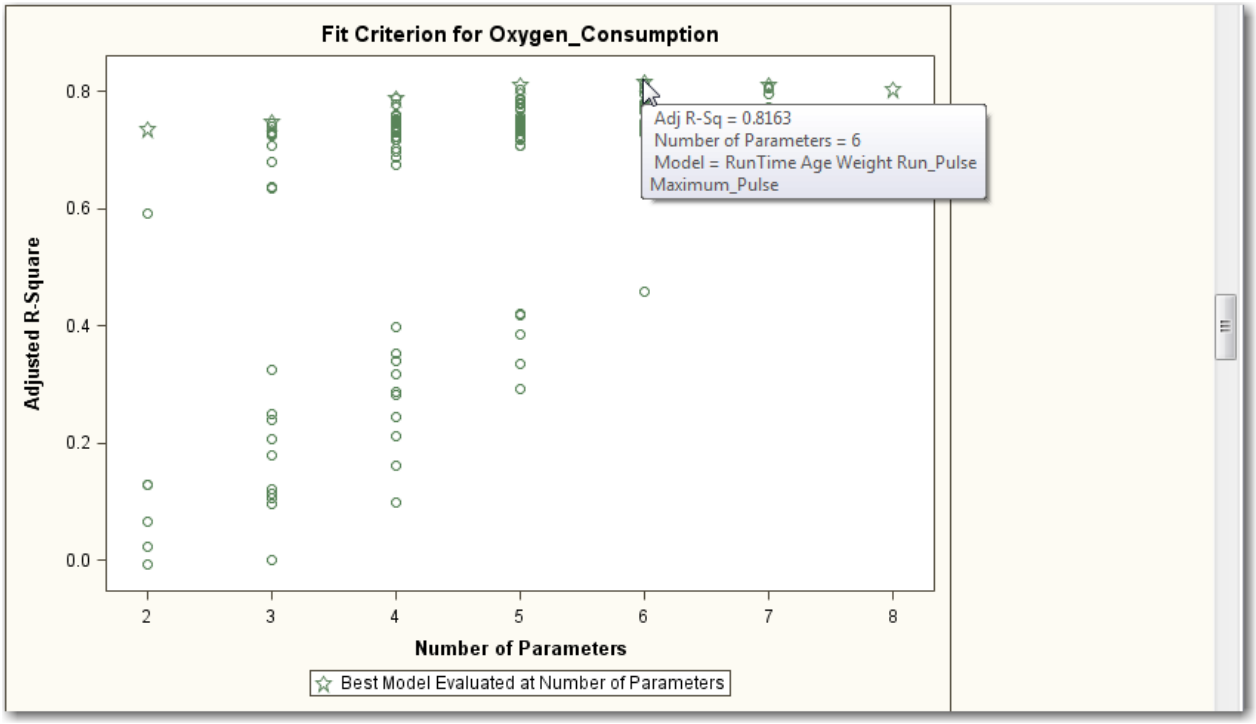


The R-Square plot compares all models based on their R^2 values. As noted earlier, adding variables to a model always increases R2, and therefore the full model is always best. Therefore, you can only use the R^2 value to compare models of equal numbers of parameters.



The adjusted R^2 does not have the problem that the R^2 has. You can compare models of different sizes. In this case, it is difficult to see which model has the higher adjusted R^2, which is the starred model for six parameters or seven parameters.

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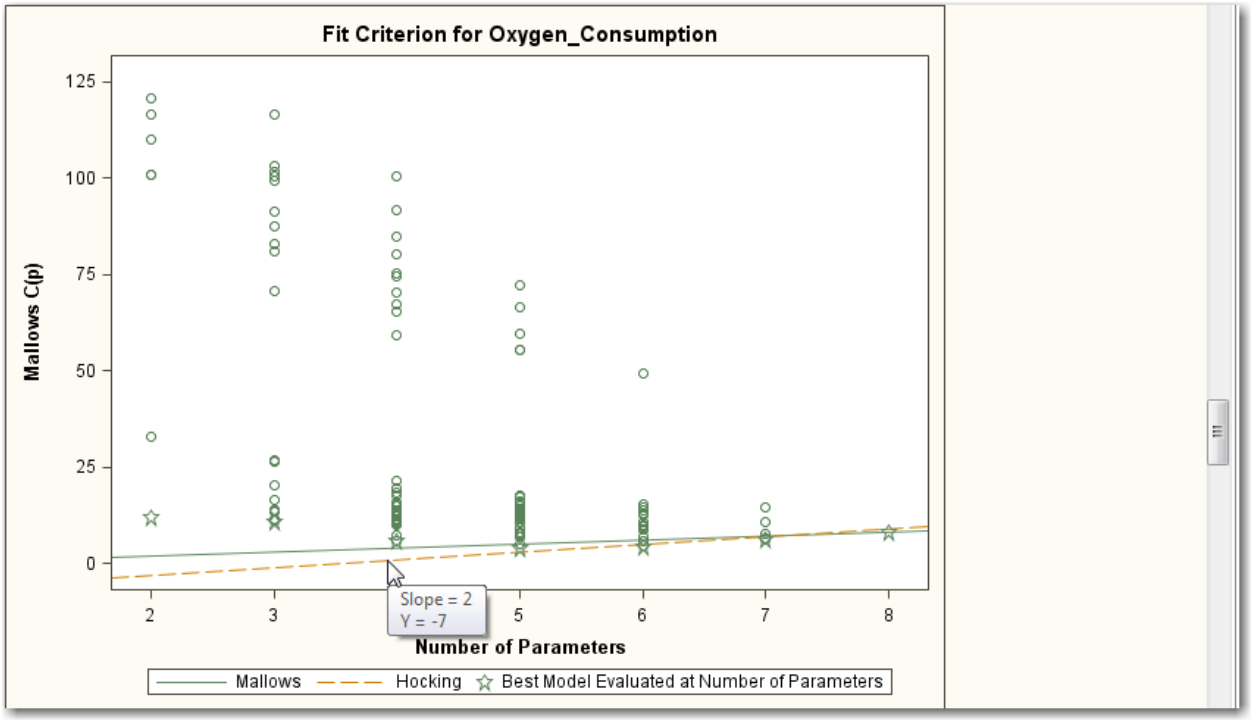

The line $C_p = p$ is plotted to help you identify models that satisfy the criterion $C_p \leq p$ for prediction. The lower line is plotted to help identify which models satisfy Hocking's criterion $C_p \leq 2p - p_{\text{full}} + 1$ for parameter estimation.

Use the graph and review the output to select a relatively short list of models that satisfy the criterion appropriate for your objective. The first model to fall below the line for Mallows' criterion has five parameters. The first model to fall below Hocking's criterion has six parameters.

It is often the case that the best model is difficult to see because of the range of C_p values at the high end. These models are clearly not the best and therefore you can focus on the models near the bottom of the range of C_p.

1. Modify the previous Linear Regression task.
2. Click **Preview code**.
3. Click **Insert Code**.

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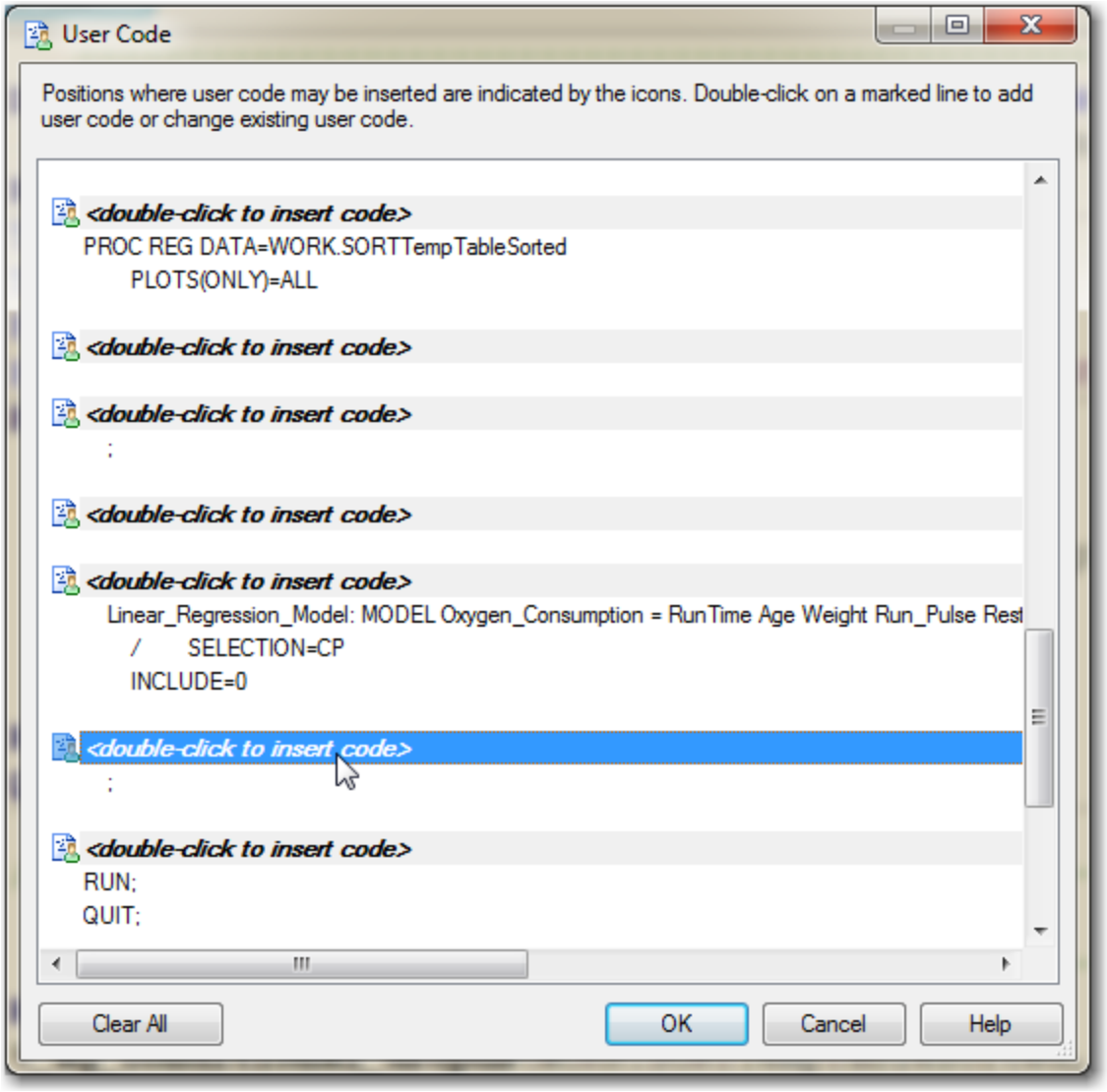
4. Double-click <***double-click to insert code***> under the INCLUDE=0 line of the code.



5. Type **BEST=20** in the Enter User Code window.

6. Click **OK** and then click **OK** to close the last two windows opened. Click in the Code Preview for Task window.

7. Click **Run**. Do not replace the results from the previous run.

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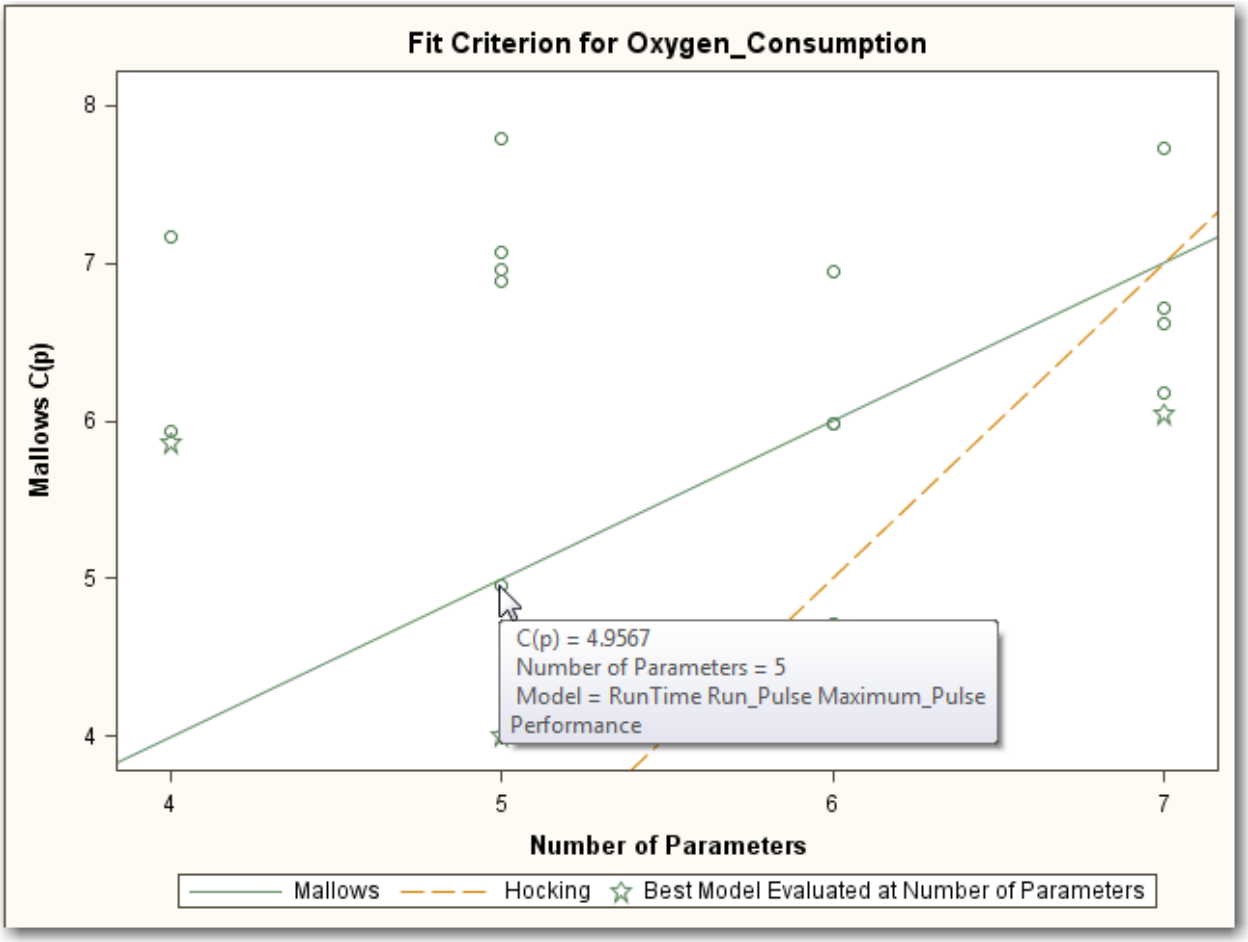
Investigate the plot of Mallows' C(p).



In this example the number of variables in the full model, p_{full}, equals 8 (seven variables plus the intercept).

The smallest model with an observation below the Mallows line has p=5 (which matches to "Number in Model" of 4 in the previous table). The model with the star at five parameters and the model above it are considered "best," based on Mallows' original criterion. The starred model has a C_p=4.004, satisfying Mallows' criterion **Runtime Age Run_Pulse Maximum_Pulse**) and the one above has a value of 4.9567 (**Performance Runtime Run_Pulse Maximum_Pulse**). The only difference between the two models is that the first includes **Age** and the second includes **Performance**. By the strictest definition, the second model should be selected, because its C_p value is closest to p.

The smallest model that falls under the Hocking line has p=6. The model with the smaller C_p value will be considered the "best" explanatory model. The table shows that the first model with p=6 is **Runtime Age Weight Run_Pulse Maximum_Pulse**, with a C_p value of 4.2598. Two other models that are also below the Hocking line are **Performance Runtime Weight Run_Pulse Maximum_Pulse** and **Performance Runtime Age Run_Pulse Maximum_Pulse**. (They are nearly on top of one another in the plot.)

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“Best” Models – Prediction

The two best candidate models based on Mallows’ original criterion includes these regressor variables:

p = 5	C_p = 4.0004 R^2=0.8355 Adj. R^2=0.8102	**RunTime, Age, Run_Pulse, Maximum_Pulse**
p = 5	C_p = 4.9567 R^2=0.8292 Adj. R^2=0.8029	**Performance, RunTime, Run_Pulse, Maximum_Pulse**

79

Some models might be essentially equivalent based on their C_p, R^2, or other measures. When, as in this case, there are several candidate “best” models, it is the responsibility of the investigator to determine which model makes the most sense based on theory and experience. The choice between these two models is essentially the choice between **Age** and **Performance**. Because age is much easier to measure than the subjective measure of performance, the first model is selected here.

A limitation of the evaluation that you did thus far is that you do not know the magnitude and signs of the coefficients of the candidate models or their statistical significance.

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“Best” Models – Parameter Estimation

The three best candidate models for analytic purposes, according to Hocking, include those listed below:

$p = 6$	$C_p = 4.2598$ $R^2=0.8469$ Adj. $R^2=0.8163$	**RunTime, Age, Weight, Run_Pulse, Maximum_Pulse**
$p = 6$	$C_p = 4.7158$ $R^2=0.8439$ Adj. $R^2=0.8127$	**Performance, RunTime, Weight, Run_Pulse, Maximum_Pulse**
$p = 6$	$C_p = 4.7168$ $R^2=0.8439$ Adj. $R^2=0.8127$	**Performance, RunTime, Age, Run_Pulse, Maximum_Pulse**

The variables **RunTime**, **Run_Pulse**, and **Maximum_Pulse** once again appear in all candidate models. The choice of models depends on the selection of pairs from **Performance**, **Age**, and **Weight**. You again choose a model with objective measures, **Age** and **Weight**. That is the top model in the list. Your choice might differ.

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Estimating and Testing the Coefficients for the Selected Models

Example: Invoke the Linear Regression task to compare the ANOVA tables and parameter estimates for the two-candidate models in the **Fitness** data set.

Optional: First, return reporting in SAS Enterprise Guide to SAS Report from HTML.

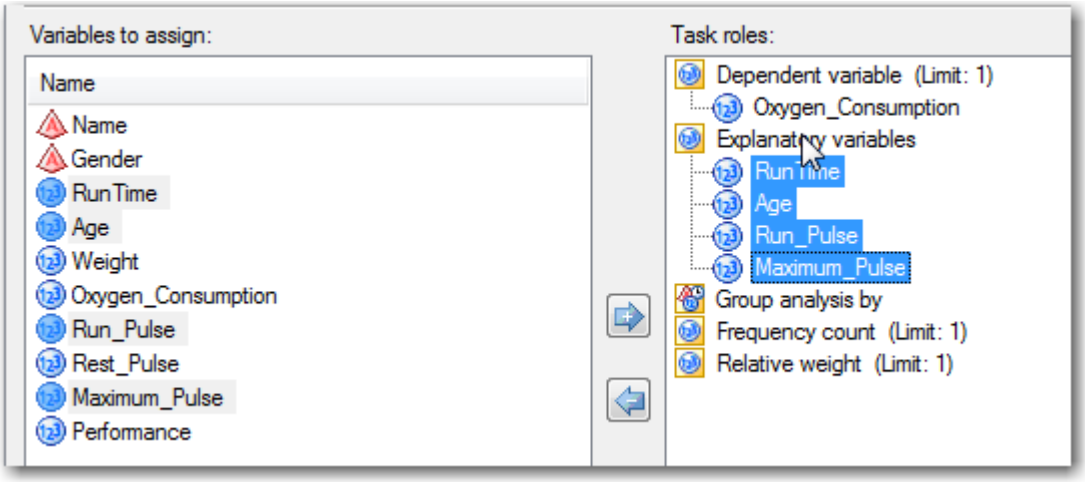
1. Select **Tools** ⇨ **Options**.

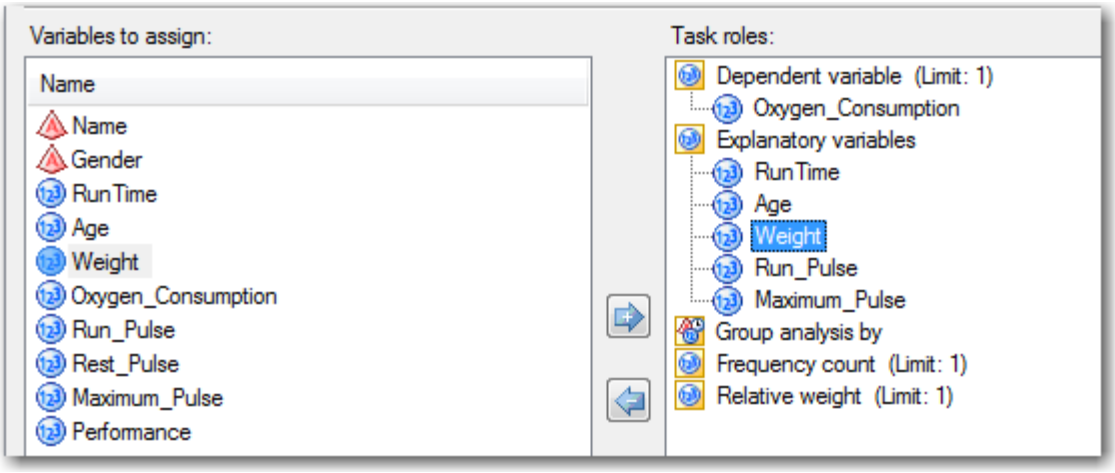
2. In the window that opens, select **Results General** under **Results** at the left and then uncheck the box for **HTML** and check the box for **SAS Report**.

3. Click **OK**.

Begin Linear Regression task.

4. Run the Linear Regression task twice, once using the variables **Runtime**, **Age**, **Run_Pulse**, and **Maximum_Pulse** as the explanatory variables and once using **Runtime**, **Age**, **Weight**, **Run_Pulse**, and **Maximum_Pulse** as the explanatory variables.





5. In each case, with **Plots** selected at the left, uncheck the box for **Show plots for regression analysis**.

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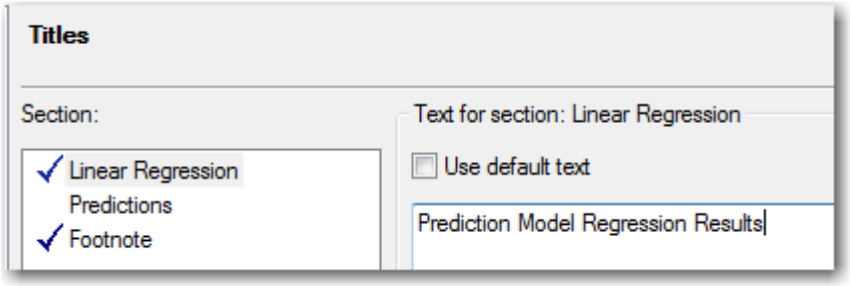
6. With **Titles** selected at the left, uncheck the box for **Use default text** and then type **Prediction Model Regression Results** in the text area for the first model and **Explanatory Model Regression Results** in the text area for the second model.



7. Click **Run**.

Output for the Prediction Model

Number of Observations Read	31
Number of Observations Used	31

Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	4	711.45087	177.86272	33.01	<.0001
Error	26	140.10368	5.38860		
Corrected Total	30	851.55455			

Root MSE	2.32134	**R-Square**	0.8355
Dependent Mean	47.37581	**Adj R-Sq**	0.8102
Coeff Var	4.89984		

Parameter Estimates					
Variable	**DF**	**Parameter Estimate**	**Standard Error**	**t Value**	**Pr > |t|**
Intercept	1	97.16952	11.65703	8.34	<.0001
RunTime	1	-2.77576	0.34159	-8.13	<.0001
Age	1	-0.18903	0.09439	-2.00	0.0557
Run_Pulse	1	-0.34568	0.11820	-2.92	0.0071
Maximum_Pulse	1	0.27188	0.13438	2.02	0.0534

The R^2 and adjusted R^2 are the same as calculated during the model selection program. If there are missing values in the data set, however, this might not be true.

The model *F* is large and highly significant. **Age** and **Maximum_Pulse** are not significant at the 0.05 level of significance. However, all terms are significant at alpha=0.10.

The adjusted R^2 is close to the R^2, which suggests that there are not too many variables in the model.

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Output for the Explanatory Model

Number of Observations Read	31
Number of Observations Used	31

Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	5	721.20532	144.24106	27.66	<.0001
Error	25	130.34923	5.21397		
Corrected Total	30	851.55455			

Root MSE	2.28341	**R-Square**	0.8469
Dependent Mean	47.37581	**Adj R-Sq**	0.8163
Coeff Var	4.81978		

Parameter Estimates					
Variable	**DF**	**Parameter Estimate**	**Standard Error**	**t Value**	**Pr > |t|**
Intercept	1	101.33835	11.86474	8.54	<.0001
RunTime	1	-2.68846	0.34202	-7.86	<.0001
Age	1	-0.21217	0.09437	-2.25	0.0336
Weight	1	-0.07332	0.05360	-1.37	0.1836
Run_Pulse	1	-0.37071	0.11770	-3.15	0.0042
Maximum_Pulse	1	0.30603	0.13452	2.28	0.0317

The adjusted R^2 is slightly larger than in the Prediction model and very close to the R^2.

The model *F* is large, but smaller than in the Prediction model. However, it is still highly significant. All terms included in the model are significant except **Weight**. The *p*-values for **Age**, **Run_Pulse**, and **Maximum_Pulse** are smaller in this model than they were in the Prediction model.

Including the additional variable in the model changes the coefficients of the other terms and changes the *t* statistics for all.

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Stepwise Regression

Example: Select a model for predicting **Oxygen_Consumption** in the **Fitness** data set by using the forward selection, backward elimination, and stepwise selection methods.

1. With the **Fitness** data set selected, select **Tasks** ⇨ **Regression** ⇨ **Linear Regression**.
2. Drag **Oxygen_Consumption** to the dependent variable task role and all other numeric variables to the explanatory variables task role.
3. With **Model** selected at the left, find the pull-down menu for **Model selection method** and click to find **Forward Selection** at the bottom.
4. With **Titles** selected at the left, deselect the box for **Use default text** and then type **Forward Selection Results** in the text area.
5. Click **Run**.

Partial Output

Number of Observations Read	31
Number of Observations Used	31

Forward Selection: Step 1

Variable RunTime Entered: R-Square = 0.7434 and C(p) = 11.9967

Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	1	633.01458	633.01458	84.00	<.0001
Error	29	218.53997	7.53586		
Corrected Total	30	851.55455			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	82.42494	3.85582	3443.63138	456.97	<.0001
RunTime	-3.31085	0.36124	633.01458	84.00	<.0001

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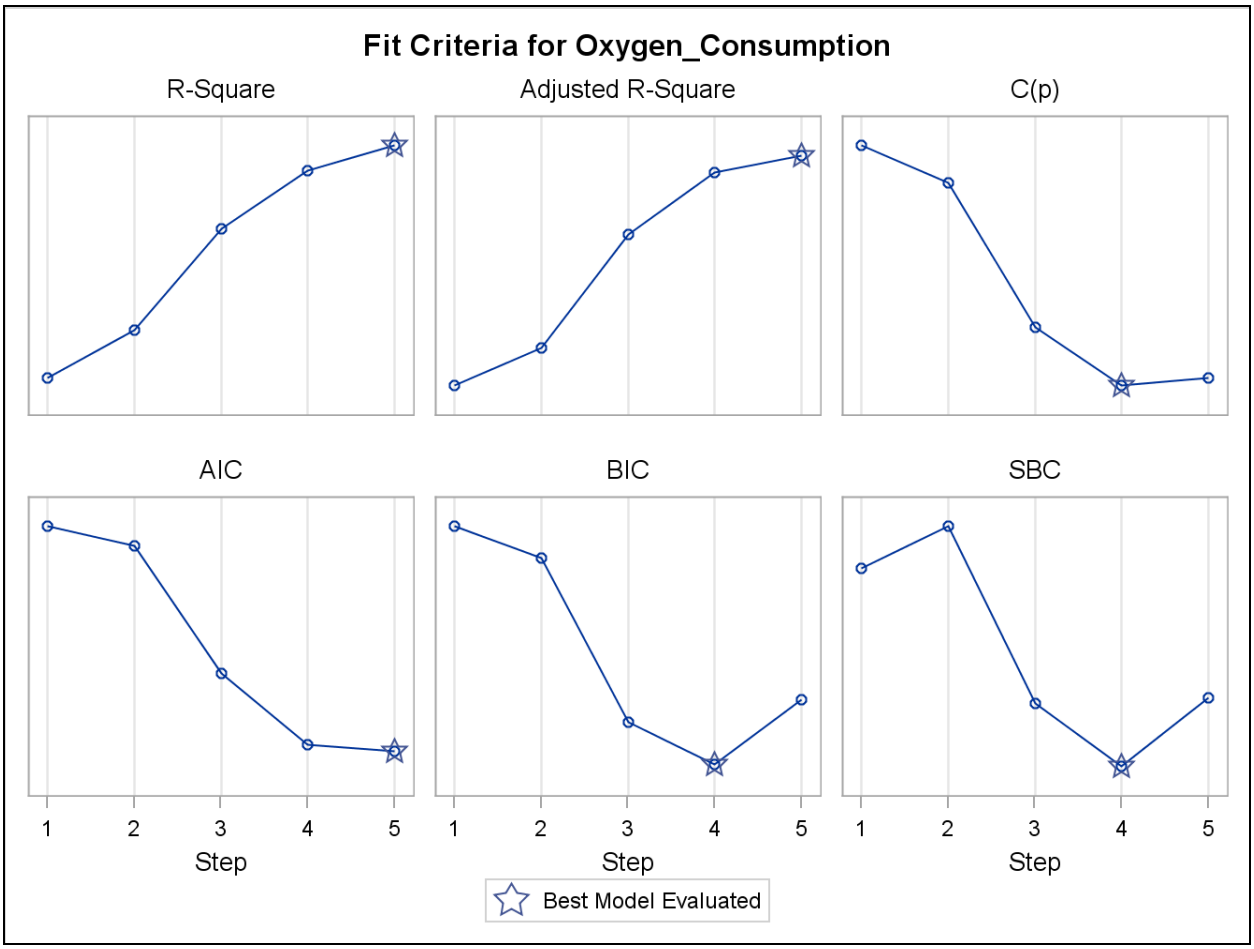
Partial Output (Continued)

Summary of Forward Selection							
Step	Variable Entered	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	RunTime	1	0.7434	0.7434	11.9967	84.00	<.0001
2	Age	2	0.0213	0.7647	10.7530	2.54	0.1222
3	Run_Pulse	3	0.0449	0.8096	5.9367	6.36	0.0179
4	Maximum_Pulse	4	0.0259	0.8355	4.0004	4.09	0.0534
5	Weight	5	0.0115	0.8469	4.2598	1.87	0.1836

The model selected at each step is printed and a summary of the sequence of steps is given at the end of the output. In the summary, the variables are listed in the order in which they were selected. The partial R^2 shows the increase in the model R^2 as each term was added.

The model selected has the same variables as the model chosen using the all-regressions techniques with the Hocking criterion. This will not always be the case.



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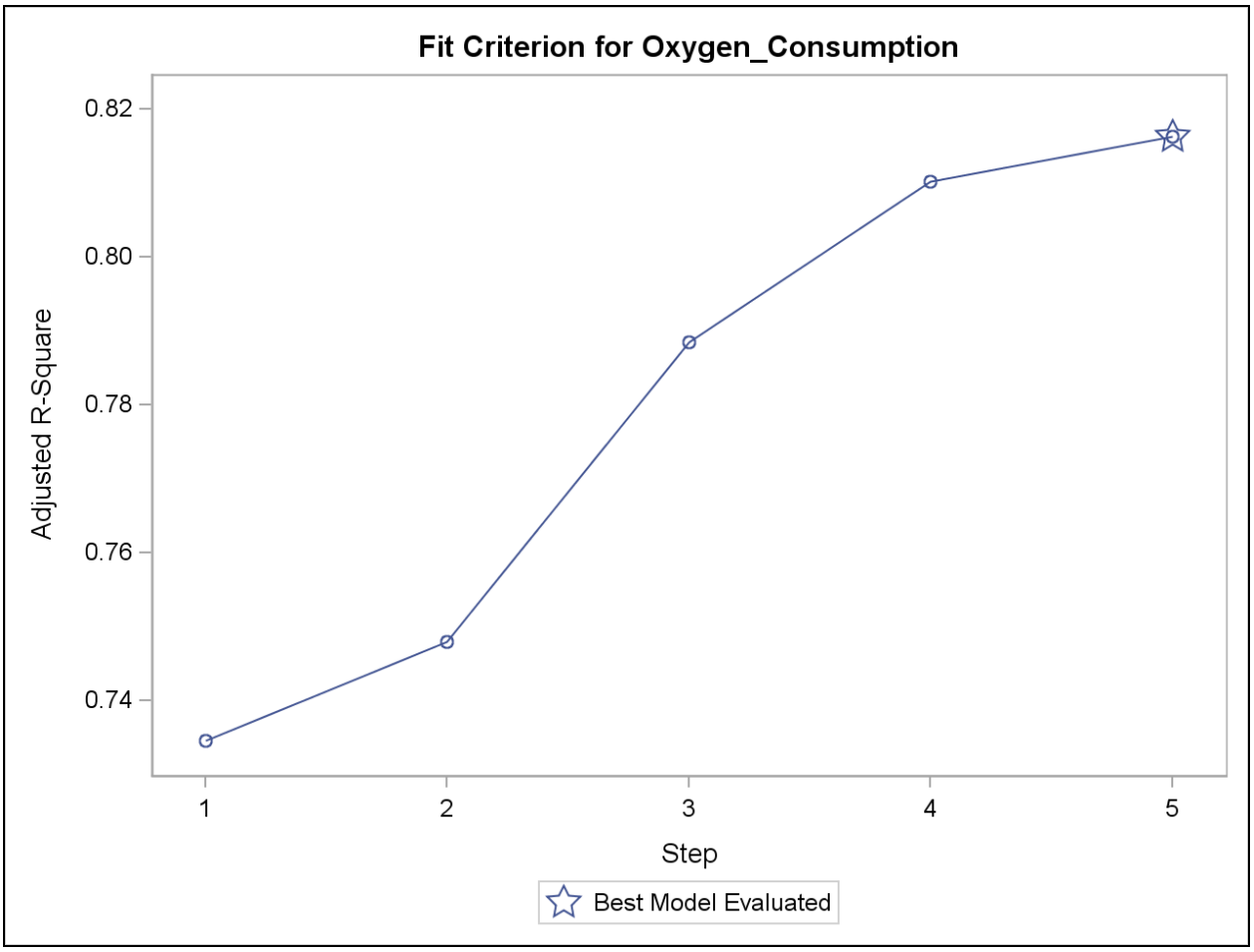

The Adjusted R-Square plot shows the progression of that statistic at each step. The star denotes the best model of the five that were tested. This is not necessarily the highest adjusted R^2 value of all possible subsets, but is the best of the five tested in the Forward model.

For backward elimination:

6. Modify the previous Linear Regression task.

7. With **Model** selected at the left, find the pull-down menu for **Model selection method** and click to find **Backward Elimination** at the bottom.

8. With **Titles** selected at the left, deselect the box for **Use default text** and then type **Backward Elimination Results** in the text area.

9. Click **Run**. Do not replace the results from the previous run.

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Partial Output

Backward Elimination: Step 0

All Variables Entered: R-Square = 0.8486 and C(p) = 8.0000

Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	7	722.66124	103.23732	18.42	<.0001
Error	23	128.89331	5.60406		
Corrected Total	30	851.55455			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	131.78249	72.20754	18.66607	3.33	0.0810
RunTime	-3.86019	2.93659	9.68350	1.73	0.2016
Age	-0.46082	0.58660	3.45842	0.62	0.4401
Weight	-0.05812	0.06892	3.98514	0.71	0.4078
Run_Pulse	-0.36207	0.12324	48.37354	8.63	0.0074
Rest_Pulse	-0.01512	0.06817	0.27581	0.05	0.8264
Maximum_Pulse	0.30102	0.13981	25.97886	4.64	0.0420
Performance	-0.12619	0.30097	0.98519	0.18	0.6789

Bounds on condition number: 162.85, 2262.9

Backward Elimination: Step 1

Variable Rest_Pulse Removed: R-Square = 0.8483 and C(p) = 6.0492

Analysis of Variance					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Model	6	722.38543	120.39757	22.37	<.0001
Error	24	129.16912	5.38205		
Corrected Total	30	851.55455			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	133.73795	70.23358	19.51494	3.63	0.0689
RunTime	-3.99624	2.81438	10.85139	2.02	0.1685
Age	-0.47577	0.57106	3.73583	0.69	0.4130
Weight	-0.05545	0.06650	3.74132	0.70	0.4126
Run_Pulse	-0.36430	0.12037	49.29878	9.16	0.0058
Maximum_Pulse	0.30184	0.13696	26.13890	4.86	0.0374
Performance	-0.13647	0.29144	1.18011	0.22	0.6438

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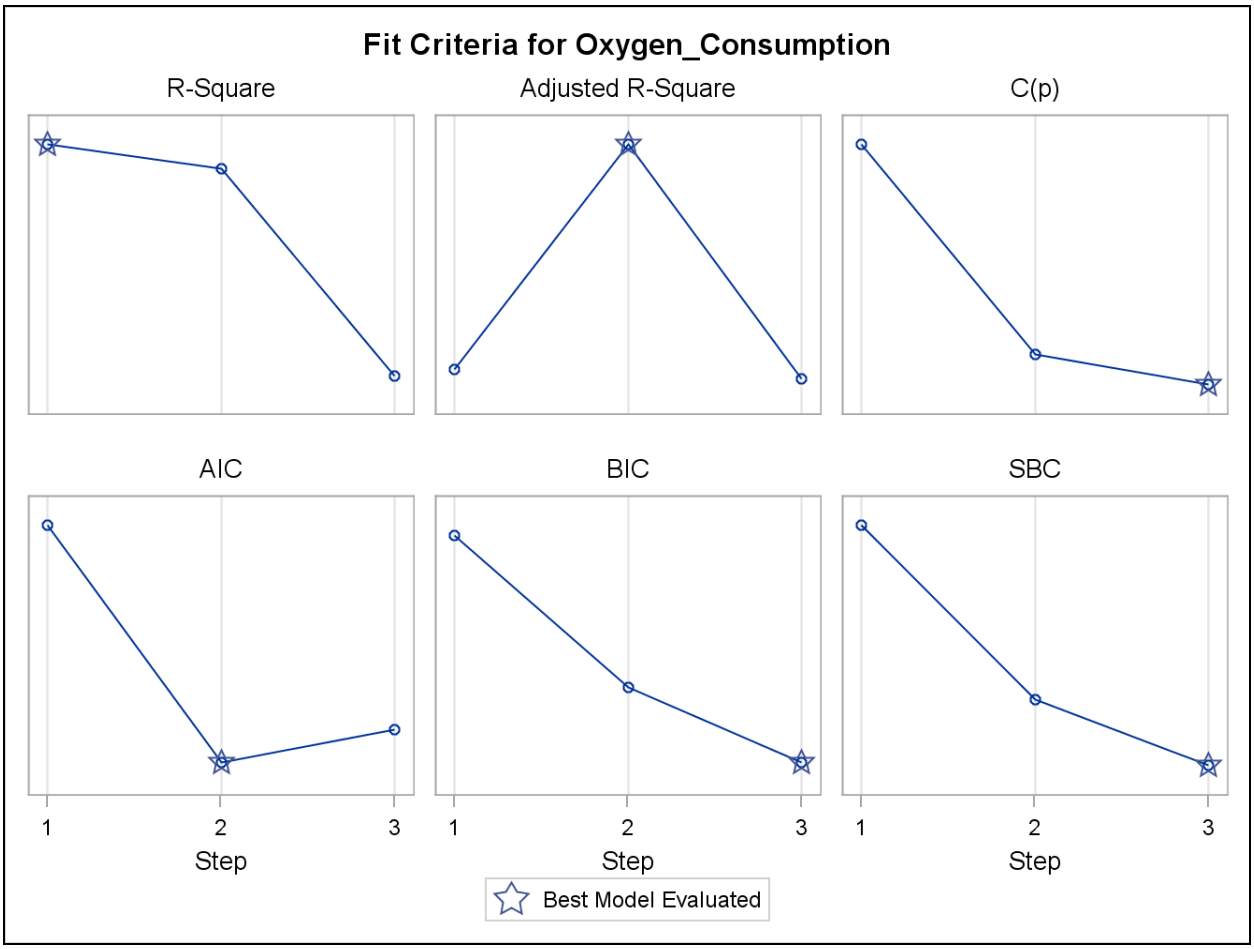
Partial Output (Continued)

Summary of Backward Elimination							
Step	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	Rest_Pulse	6	0.0003	0.8483	6.0492	0.05	0.8264
2	Performance	5	0.0014	0.8469	4.2598	0.22	0.6438
3	Weight	4	0.0115	0.8355	4.0004	1.87	0.1836

Using the Backward Elimination option and the default *p*-value, three independent variables were eliminated. By coincidence the final model is the same as the one considered best based on C_p, using the Mallows criterion.



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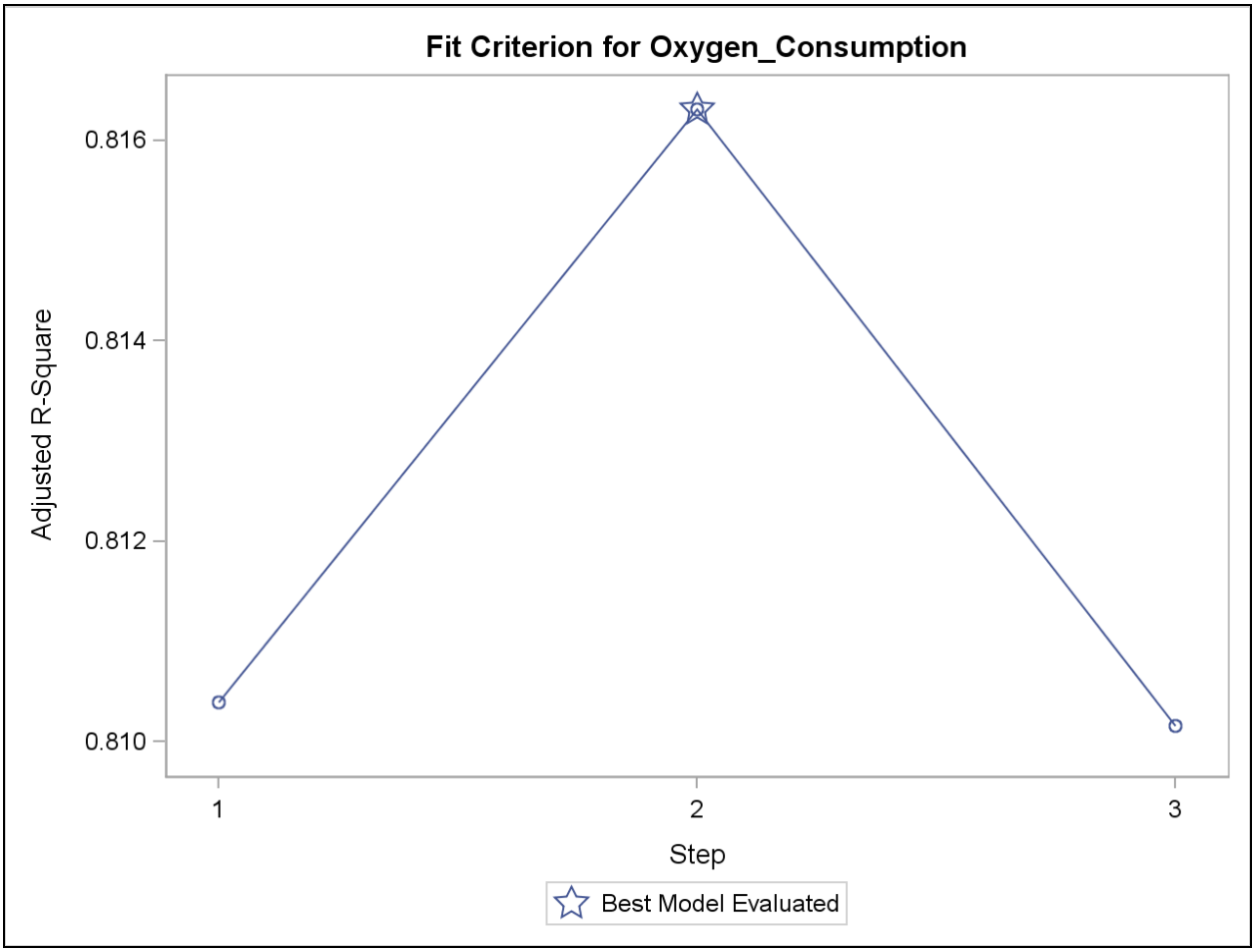

The adjusted R^2 for the model at step 2 (before **Weight** was removed) was greatest of the three tested.

For Stepwise Selection:

10. Modify the previous Linear Regression task.
11. With **Model** selected at the left, find the pull-down menu for **`Model selection method`** and click to find **Stepwise Selection** at the bottom.
12. With **Titles** selected at the left, deselect the box for **Use default text** and then type **Stepwise Selection Results** in the text area.
13. Click **Run**. Do not replace the results from the previous run.

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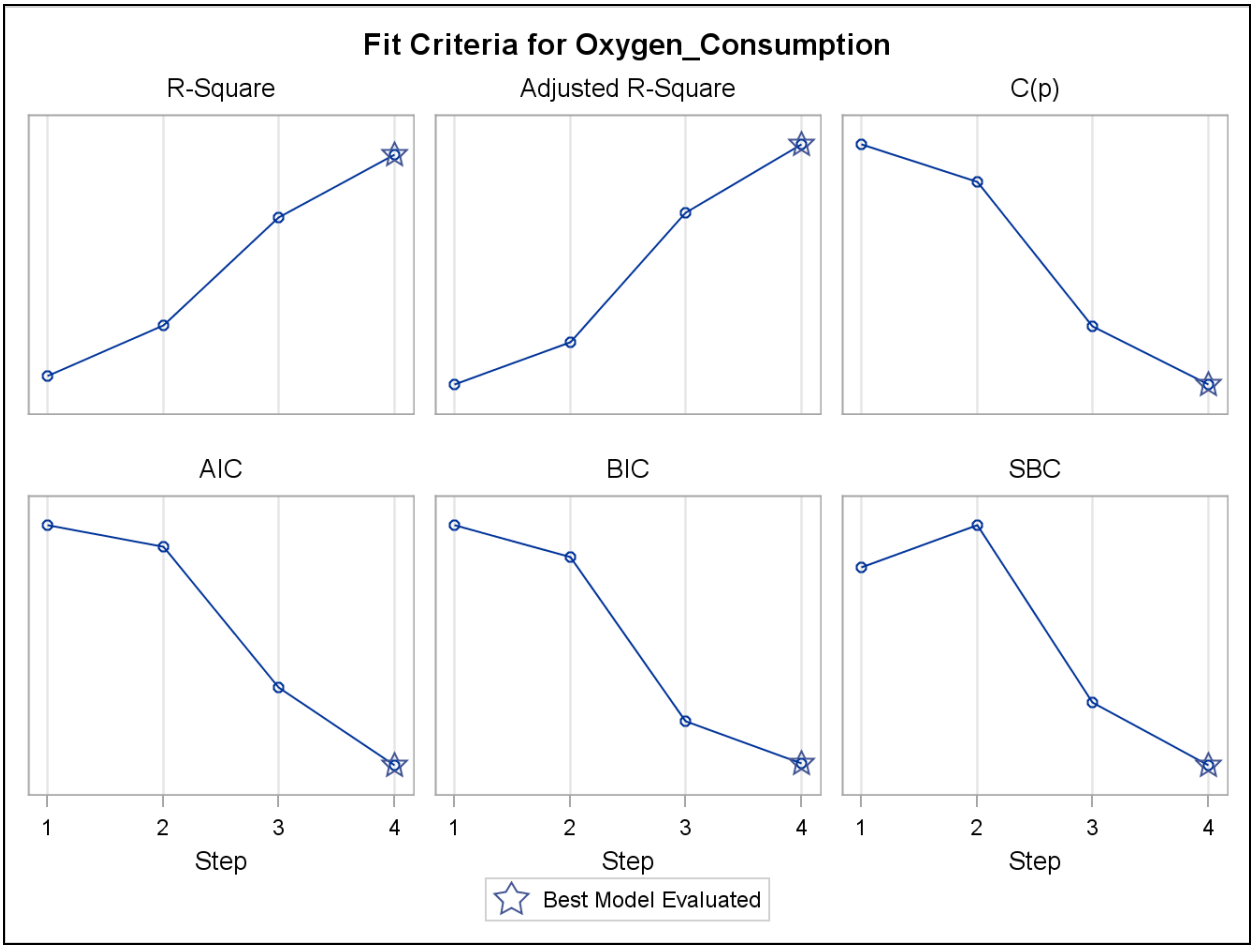
Partial Output

Summary of Stepwise Selection								
Step	Variable Entered	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	RunTime		1	0.7434	0.7434	11.9967	84.00	<.0001
2	Age		2	0.0213	0.7647	10.7530	2.54	0.1222
3	Run_Pulse		3	0.0449	0.8096	5.9367	6.36	0.0179
4	Maximum_Pulse		4	0.0259	0.8355	4.0004	4.09	0.0534

Using Stepwise Selection and the default variable entry and removal *p*-value criteria, the same subset resulted as that using the Backward Elimination option.



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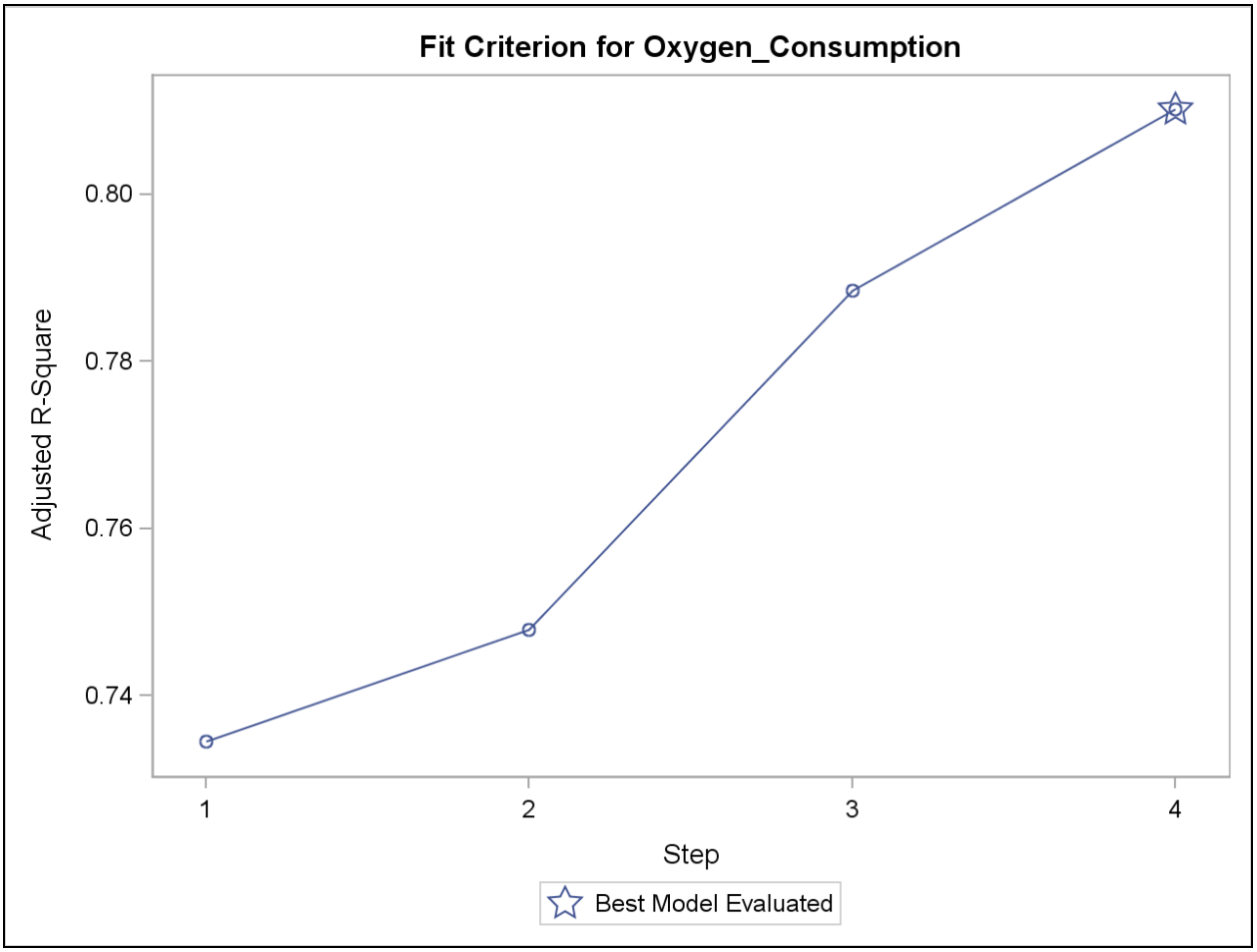

The default significance level criterion is $p<.50$ for the Forward Selection method and $p<.15$ for the Stepwise Selection method. After **RunTime** was entered into the model, **Age** was entered at step 2 with a *p*-value of 0.1222. If the significance level criterion were set to something less than 0.10, the final model would be quite different. It would include only one variable, **RunTime**. This underscores the precariousness of relying on one stepwise method for defining a "best" model.

The scale of the default Y axes in these plots might give misleading information about the effect of adding or removing variables. The same plots displayed side-by-side and using a common y-scale of 0 to 1 is shown below. The differences do not look nearly as great.

The "Bounds on the condition number" reported at each step of the output for the selection methods refer to a measurement of *collinearity* (correlation among predictor variables).

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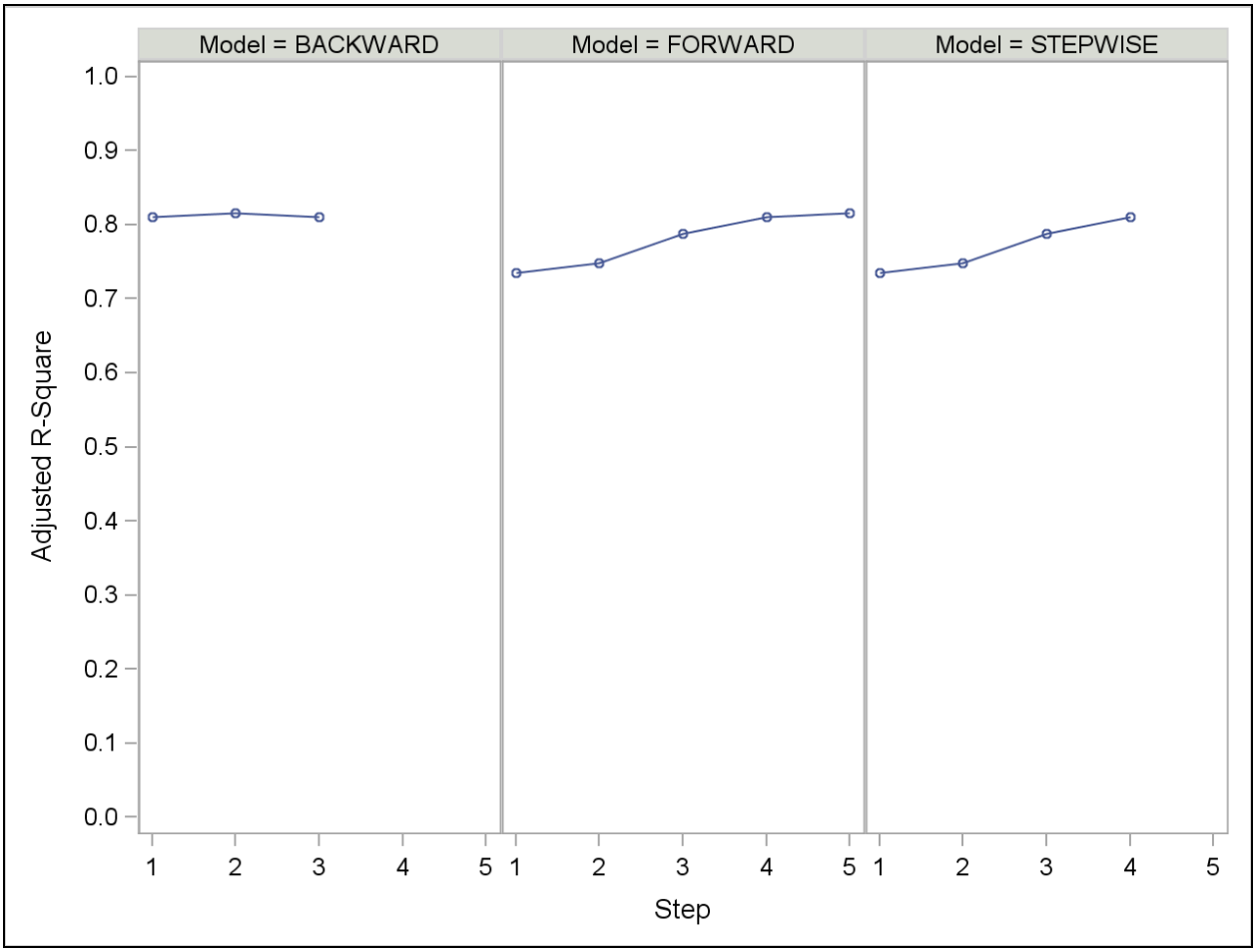

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Stepwise Regression Models*

Forward Selection	**Runtime**, **Age**, **Weight**, **Run_Pulse**, **Maximum_Pulse**
Backward Elimination	**Runtime**, **Age**, **Run_Pulse**, **Maximum_Pulse**
Stepwise Selection	**Runtime**, **Age**, **Run_Pulse**, **Maximum_Pulse**

*** Using default significance levels**

111

The final models obtained using the default entry and stay significance level criteria are displayed. It is important to note that the choice of criterion levels can greatly affect the final models that are selected using stepwise methods.

Stepwise Regression Models#

Forward Selection	**Runtime**
Backward Elimination	**Runtime**, **Run_Pulse**, **Maximum_Pulse**
Stepwise Selection	**Runtime**

 # Using 0.05 significance levels

112

The final models using 0.05 as the forward and backward step criteria resulted in very different models than those chosen using the default criteria.

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